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Z → τ_hτ_h CROSS SECTION WITH CMS OPEN DATA

v3: DeepTau Tight on both legs, MC-subtracted closure-corrected fake factor, fiducial signal, k-fold BDT categories. 15 September 2026

$\sigma(pp \rightarrow Z/\gamma^* \rightarrow \tau\tau, 60 < m < 120 \text{ GeV}) = 2082 \pm 41 \text{ (stat)} \pm_{194}^{222} \text{ (syst)} \pm 76 \text{ (acc) pb} \quad (\text{NNLO } 1945 \text{ pb})$

$\sigma_{\text{fid}}(\tau_h\tau_h) = 4.82 \pm 0.09 \pm 0.47 \text{ pb} \quad (\text{pred. } 4.50 \text{ pb}), \quad \mu_Z = 1.071 \pm_{0.100}^{0.114} \quad (\text{v2.1 Medium: } 1.205 +0.145 -0.125; \text{v1: } 1.160 +0.192 -0.163)$

CMS 2016 Tau dataset, Run2016G+H, 16.4 fb-1; both taus hadronic, DeepTau Tight; fakes from data (same-sign fake factor); TRExFitter v1.8.0.

Precision +11/-10 % (v2.1 Medium: +14/-12 %, v1: +19/-16 %): tau ID SFs 8 %, fakes 7 %, MC stat 4 %, trigger 3 %; data statistics 2.1 %; GoF p = 0.25.

Sections: 1 result 2 what changed (v1 -> v2 -> v2.1 -> v3) 3 fit 4 BDT categories 5 fakes 6 signal definition and selection 7 findings, next steps

z-tautau/: README.md, CLAUDE.md, REVIEW.md, docs/00-09, review/ (code, documentation, review studies)

1. RESULT

NanoAOD: 160 GB data, 4 Drell-Yan + background samples



skims (trigger, ≥ 2 τ_h candidates): 1.7 GB



ntuples: pair, trigger match, vetoes, masses: 330 MB (laptop)



fake factors (era $\times$ DM $\times$ N_{jets} $\times$ p_T), MC subtracted,
closure corrections $f(|\eta_1|) g(p_{T,2})$, $C_{\text{OS/SS}}$



k-fold BDT (5 folds by event number) $\rightarrow$ SR0 / SR1 / SR2



templates: fiducial signal, non-fiducial DY as background,
39 nuisance parameters + 42 γ



TRExFitter v1.8.0: fit of $m_{\tau\tau}$ in 3 categories $\rightarrow \mu_Z, \sigma$

Nominal (v3, DeepTau Tight)

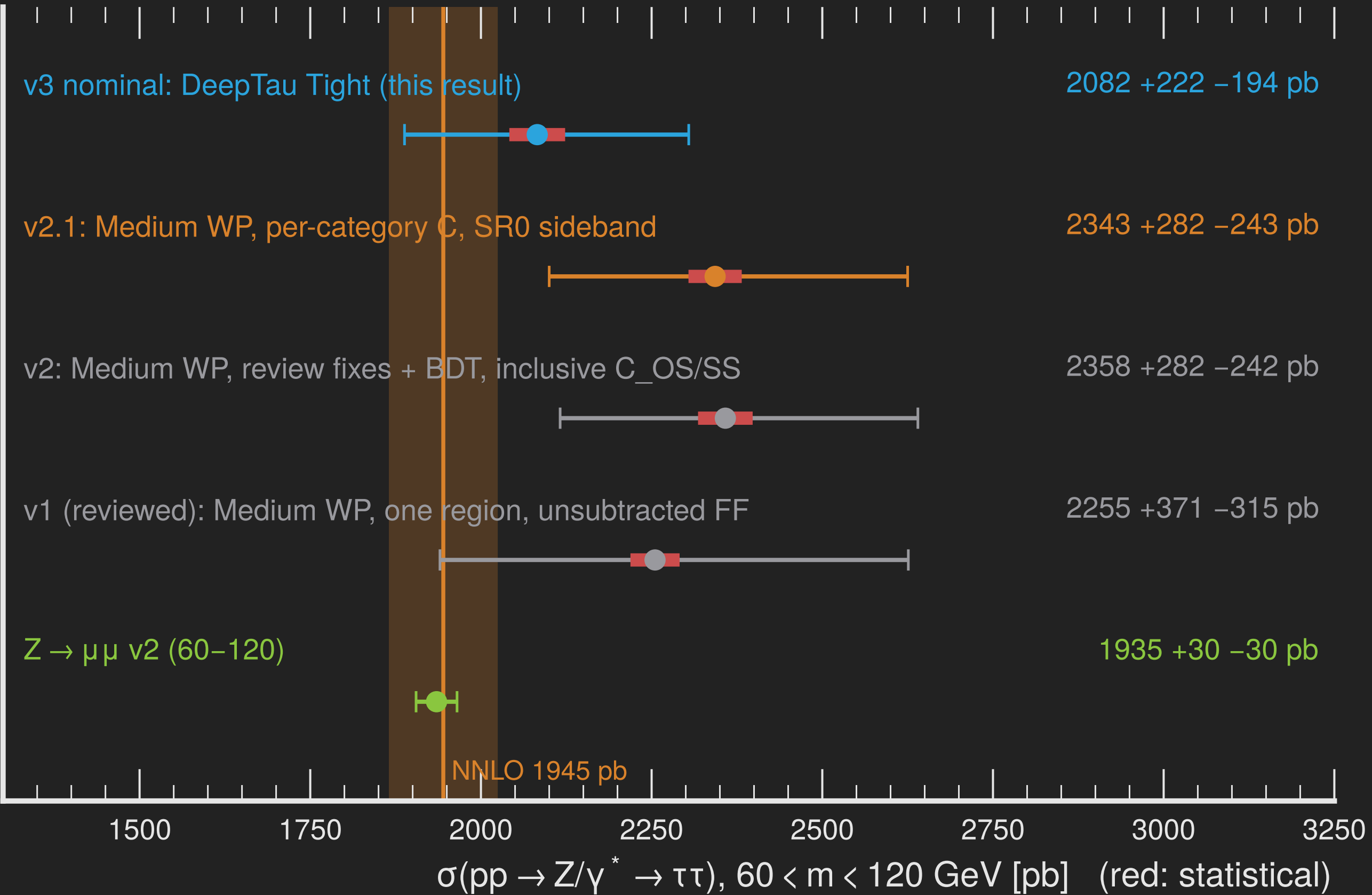
$\mu_Z = 1.071^{+0.114}_{-0.100}$ (stat. ± 0.021 , expected $^{+0.106}_{-0.093}$)
 $\sigma(60-120) = 2082^{+222}_{-194} \pm 76_A$ pb, NNLO 1945 pb
 $\sigma_{fid} = 4.82 \pm 0.09 \pm 0.47$ pb, prediction 4.50 pb
goodness of fit (saturated model): $p = 0.250$

Previous versions (same data)

v2.1, DeepTau Medium, same chain: $\mu_Z = 1.205^{+0.145}_{-0.125}$, $\sigma = 2343$ pb
v1 (single region, unsubtracted FF, whole DY as signal): $\mu_Z = 1.160^{+0.192}_{-0.163}$
 $Z \rightarrow \mu\mu$ v2: $\mu_Z = 0.990 \pm 0.014$, $\sigma(60-120) = 1935 \pm 30$ pb

Reading

0.7σ above NNLO and $\mu\mu$; a residual rise of data/prediction with the visible- τ p_T remains (section 7).
Precision limited by the external tau ID scale factors, then by the fake estimate.



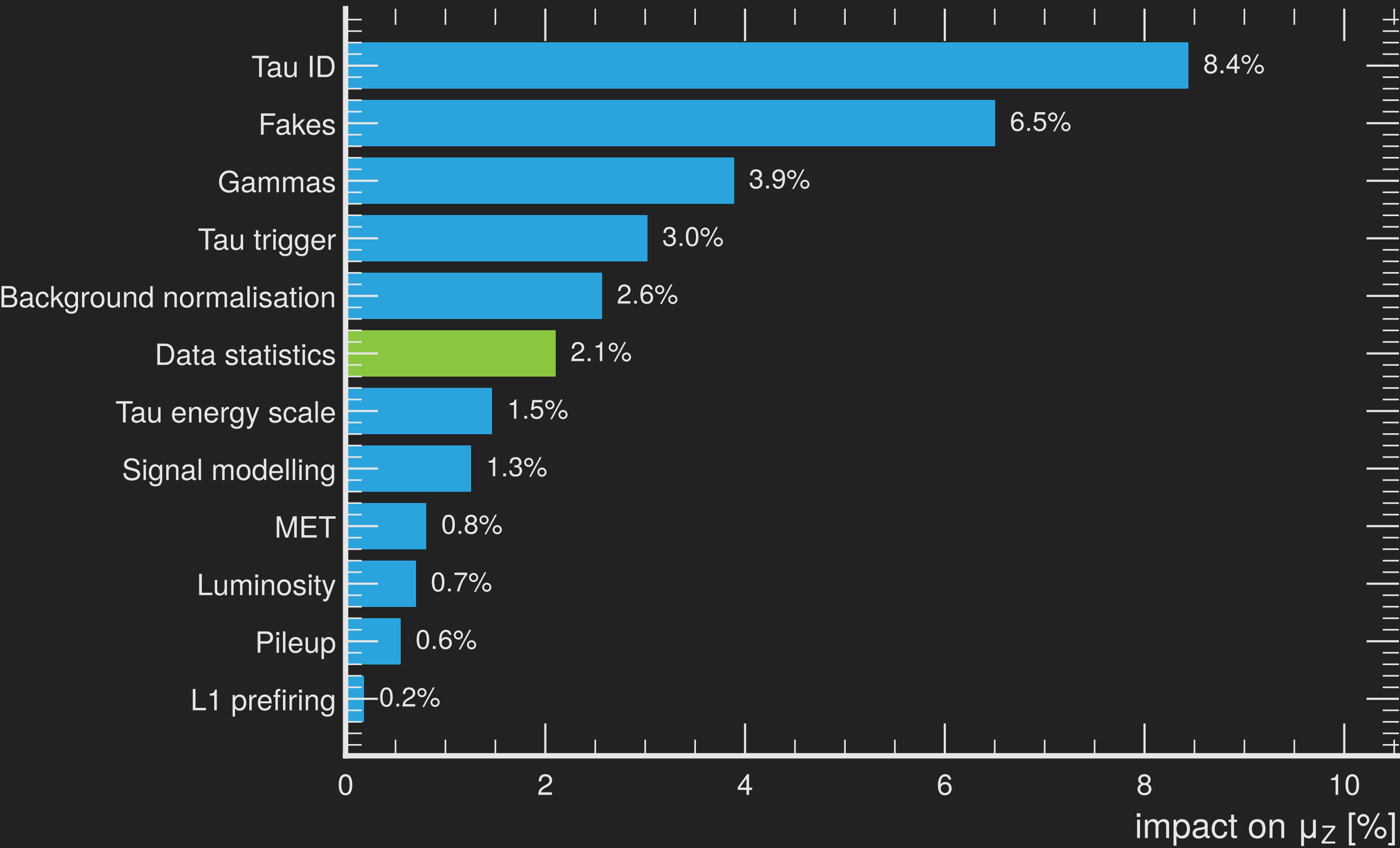
Grouped impacts on

- μ_z
- Tau ID: 8.4 %
- Fakes: 6.5 %
- Gammas: 3.9 %
- Tau trigger: 3.0 %
- Background normalisation: 2.6 %
- Tau energy scale: 1.5 %
- Signal modelling: 1.3 %
- MET: 0.8 %
- Luminosity: 0.7 %
- Pileup: 0.6 %
- L1 prefiring: 0.2 %
- data statistics: 2.1 %

v2.1 (Medium) for comparison

tau ID 11.1 %, fakes 8.6 %, MC stat 4.7 %, trigger 4.2 %; **v1**: tau ID 12.5 %, LO-vs-NLO NP 9.8 % (removed), trigger 5.1 %, MC stat 4.8 %, fakes 0.8 % (under-estimated).

Fakes carry an honest per-category closure uncertainty; MC stat improved by the jet-binned DY samples; signal modelling is scales + PS + PDF only; Tight shrinks every group.



2. WHAT CHANGED: V1 -> V2 -> V2.1 -> V3

REVIEW.md findings (v1) and what v2 does

finding	v2	effect
3.1 38 % of the signal template non-fiducial	fiducial part = signal (μ_Z); rest = DYtautau_nonfid background (5 % + scales/PDF/PS on the ratio)	definition
3.2 FF without MC subtraction (bias -0.05)	MC-subtracted FF nominal; W+jets with uniform weights	+0.05 on μ
3.4 SigModel LO vs NLO 7 %, correlated with mumu	fiducial $C(\text{LO})/C(\text{NLO}) = 0.841$ reported, not fitted; NLO scales/PS/PDF	-10 % syst
3.5 one over-constrained closure NP, FF stat coherent	closure corrections $\eta(\tau_1)$, $p_T(\tau_2)$; FF stat per event; closure NPs per category x mass region	honest fakes
3.6 MC statistics 4.8 %	DY 0J/1J/2J stitched per LHE_NpNLO (5x signal MC)	3.9 %
3.8 trigger SF on jet legs	SF = 1 for genPartFlav = 0 legs	< 1 %
3.9 second TRExFitter build (StatAnalysis ROOT, v1.10)	submodule reset to v1.8.0, rebuilt with fitting/build_trexfitter.sh	reproducible
4.3 80 % fakes: ML?	5-fold XGBoost, mass-agnostic inputs, 3 categories, m_{tautau} fitted in each	fakes where fakes are
v2.1 fake NPs pulled 0.6-1.5 sigma	$C_{\text{OS/SS}}$ per (era, N_{jets} , BDT category); SR0 fitted only above 110 GeV (fake sideband)	pulls < 0.7 sigma
4.1 Tight working point (v3)	DeepTau Tight on both legs is the nominal: 2.7x fewer fakes, every uncertainty group smaller	-2 % on the total
5 $\eta(\tau_1)$ non-closure +-15 %	diagnosed (FF depends on $ \eta $), corrected	η plots right

v1 (reviewed)

one region, unsubtracted FF, whole Z/gamma*->tautau as signal, LO-vs-NLO NP. mu_Z = 1.160 +0.192 -0.163, sigma = 2255 pb, GoF p = 0.18

v2

fiducial signal, MC-subtracted closure-corrected FF, jet-binned DY, k-fold BDT categories; one inclusive C_OS/SS. mu_Z = 1.212 +0.145 -0.125, sigma = 2358 pb, GoF p = 0.012

v2.1

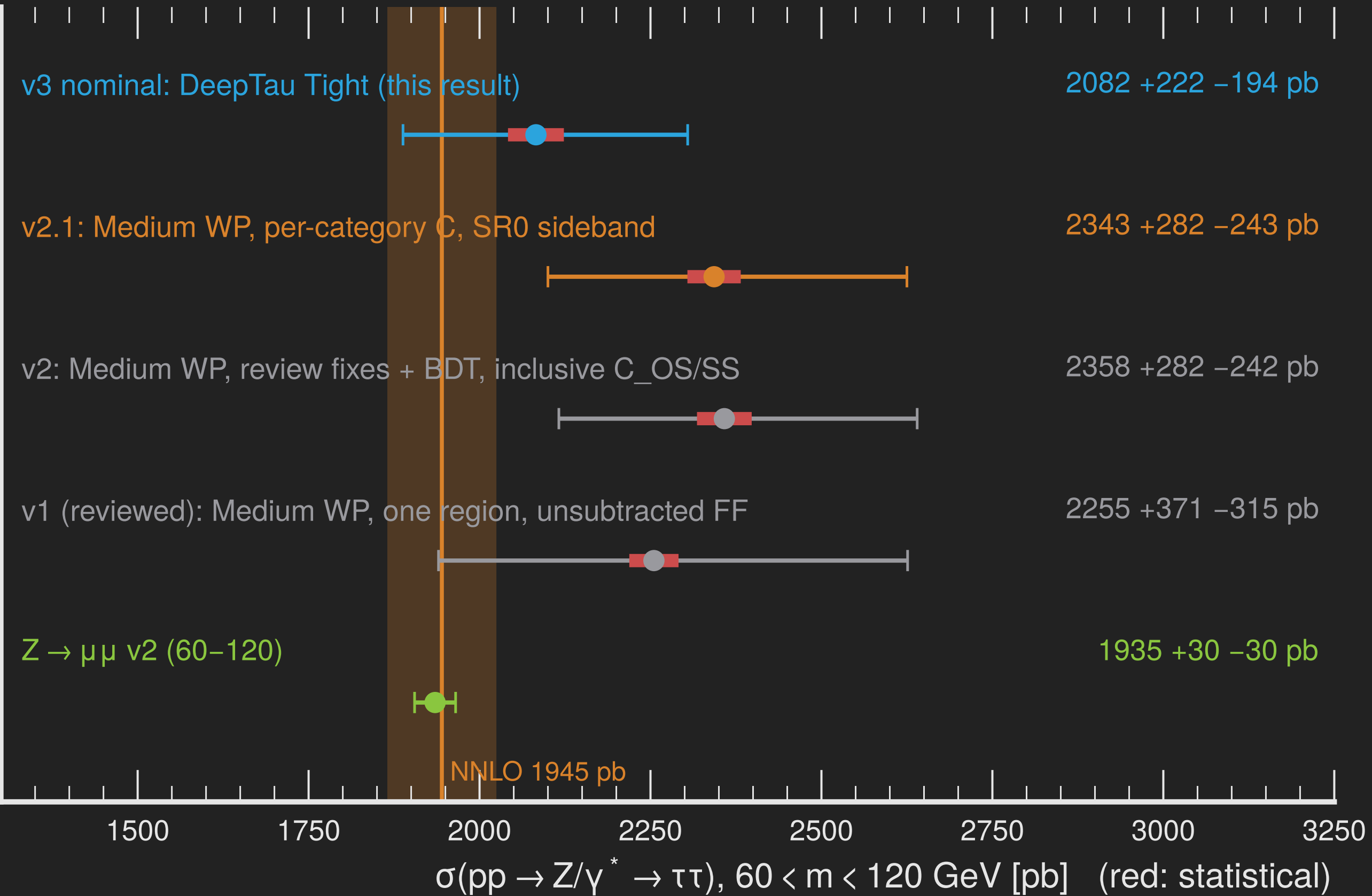
Medium WP; C_OS/SS per (era, N_jets, BDT category); fake-dominated category fitted above 110 GeV only. mu_Z = 1.205 +0.145 -0.125, sigma = 2343 pb, GoF p = 0.22

v3 (this result): DeepTau Tight on both legs

same chain, TauPOG Tight scale factors, BDT retrained. mu_Z = 1.071 +0.114 -0.100, sigma = 2082 pb, GoF p = 0.25

Reading

v1 -> v2: +0.05 from the MC subtraction, uncertainty +19/-16 % -> +14/-12 % (LO-vs-NLO NP removed, MC statistics added). v2 -> v2.1: per-category C and the SR0 sideband fixed the fake pulls and the fit quality. v2.1 -> v3: Tight removes 2.7x more fakes than signal, every uncertainty group shrinks (+11/-10 %), the central value moves 0.13 closer to NNLO.



grouped impacts; v1 'Signal modelling' was the LO-vs-NLO NP, v1 'Fakes' was under-estimated (one over-constrained NP); v1-v2.1 Medium, v3 Tight

group	v1	v2	v2.1 (Medium)	v3 (Tight)
Tau ID	12.5	11.2	11.1	8.4
Fakes	0.8	8.8	8.6	6.5
Gammas	4.8	4.7	4.7	3.9
Tau trigger	5.1	4.2	4.2	3.0
Tau energy scale	3.2	2.5	2.1	1.5
Signal modelling	9.8	1.8	1.8	1.3
Background normalisation	0.6	1.8	1.8	2.6
MET	0.6	1.2	1.0	0.8
Luminosity	1.3	1.0	1.0	0.7
Pileup	1.2	0.9	1.0	0.6
L1 prefiring	0.3	0.3	0.2	0.2
data statistics	1.8	2.0	2.0	2.1
total on mu_Z (+/-)	+19 / -16	+14 / -12	+14 / -12	+11 / -10
goodness of fit p	0.18	0.012	0.22	0.25

fake factors, closure corrections, C_OS/SS per category, BDT and fit all redone at the Tight working point with its TauPOG scale factors

quantity	Medium (v2.1)	Tight (v3)
mu_Z	1.205 +0.145 -0.125	1.071 +0.114 -0.100
total uncertainty on mu_Z	+14.5 / -12.5 %	+11.4 / -10.0 %
sigma(60-120) [pb]	2343 +282 -243	2082 +222 -194
data statistics	2.0 %	2.1 %
tau ID / fakes / MC stat / trigger	11.1 / 8.6 / 4.7 / 4.2 %	8.4 / 6.5 / 3.9 / 3.0 %
fiducial signal / fakes (prefit, all categories)	4827 / 36931	3759 / 13592
SR2: signal / fakes	2790 / 529	2141 / 197
goodness of fit p	0.22	0.25
BDT held-out AUC	0.970	0.966
tau ID SF per DM (0/1/10/11)	0.92±0.14, 0.88±0.05, 0.87±0.09, 0.90±0.16	0.90+-0.13, 0.89+-0.05, 0.94+-0.15, 0.81+-0.15

pulls (post-fit constraint) of the fake nuisance parameters

parameter	v2 (Medium)	v2.1 (Medium)	v3 (Tight)
FF OS/SS extrapolation (v2: one NP)	+0.64 (0.45)	-	-
FF OS/SS SR0 / SR1 / SR2 (per category)	-	+0.58 (0.45) / +0.34 (0.82) / +0.18 (0.97)	+0.32 (0.68) / +0.39 (0.92) / +0.25 (0.98)
FF non-closure SR1 m<110 / m>110	+0.78 (0.74) / +0.99 (0.66)	+0.11 (0.79) / +0.41 (0.70)	+0.67 (0.83) / -0.08 (0.52)
FF non-closure SR2 m<110 / m>110	-0.09 (0.96) / +1.45 (0.56)	+0.05 (0.96) / +1.44 (0.55)	+0.06 (1.01) / +1.54 (0.83)
FF non-closure SR0 m<110 / m>110	-0.05 (0.27) / -0.12 (0.95)	not fitted / +0.16 (0.96)	not fitted / +0.12 (0.95)

diagnosis (v2, Medium): tau2-anti-isolated sideband, observed / predicted with the inclusive C; and the per-category C of v3 (Tight; era G: 0 / 1 / >= 2 jets; era H)

BDT category	C used in v2	sideband obs / pred (v2)	C v3, Run2016G	C v3, Run2016H
category 0	1.052 (inclusive)	0.996	1.056 / 1.079 / 1.059	1.039 / 1.029 / 1.058
category 1	1.052 (inclusive)	1.065	1.109 / 1.172 / 1.097	1.163 / 1.149 / 1.113
category 2	1.052 (inclusive)	1.031	1.187 / 1.045 / 1.305	1.072 / 0.960 / 1.151

Reviewed (v1 -> v2)

- 38 % of the signal template non-fiducial -> fiducial signal + non-fiducial DY background (5 % NP, theory variations on the ratio)
- fake factor without MC subtraction (bias -0.05) -> MC-subtracted, W+jets with uniform weights; closure corrections in $|\eta(\tau_1)|$ (FF varies by +-15 %) and $p_T(\tau_2)$
- one over-constrained closure NP, FF statistics as 4 coherent NPs -> per-event FF statistics in the template variance, closure NPs per category and mass region
- LO-vs-NLO 'SigModel' (7 %) correlated with $m_{\mu\mu}$ -> reported (fiducial C ratio 0.867), not fitted; NLO scales, PS, PDF
- MC statistics 4.8 % -> DY 0J/1J/2J stitched per LHE_NpNLO (5x signal MC)
- 80 % fakes -> 5-fold XGBoost on mass-agnostic inputs, three categories, $m_{\tau\tau}$ fitted in each; FF closure in the score verified
- wrong TRExFitter build (v1.10, StatAnalysis ROOT) -> submodule at v1.8.0, rebuilt against the LCG ROOT

Follow-up (v2 -> v2.1 -> v3)

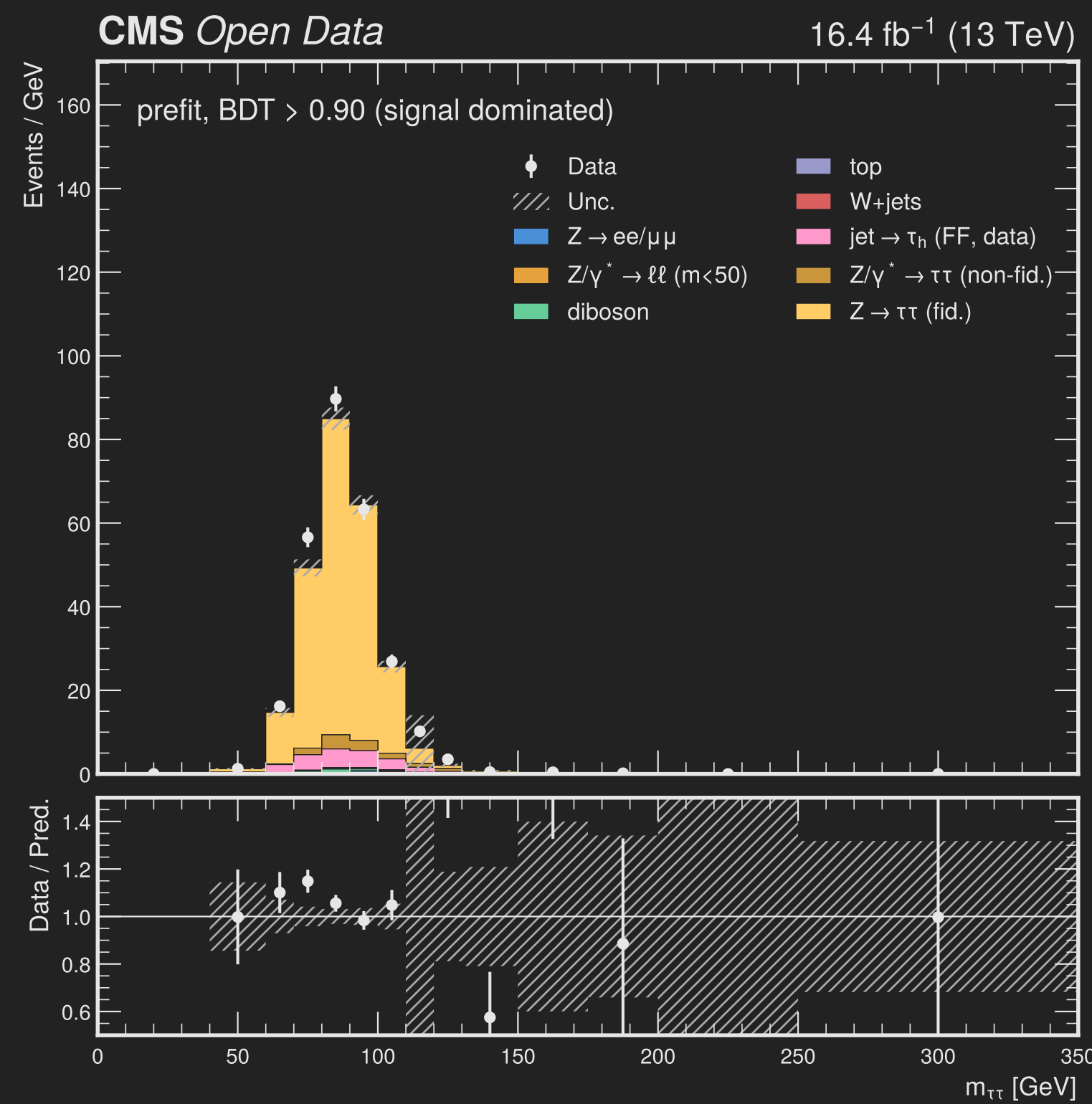
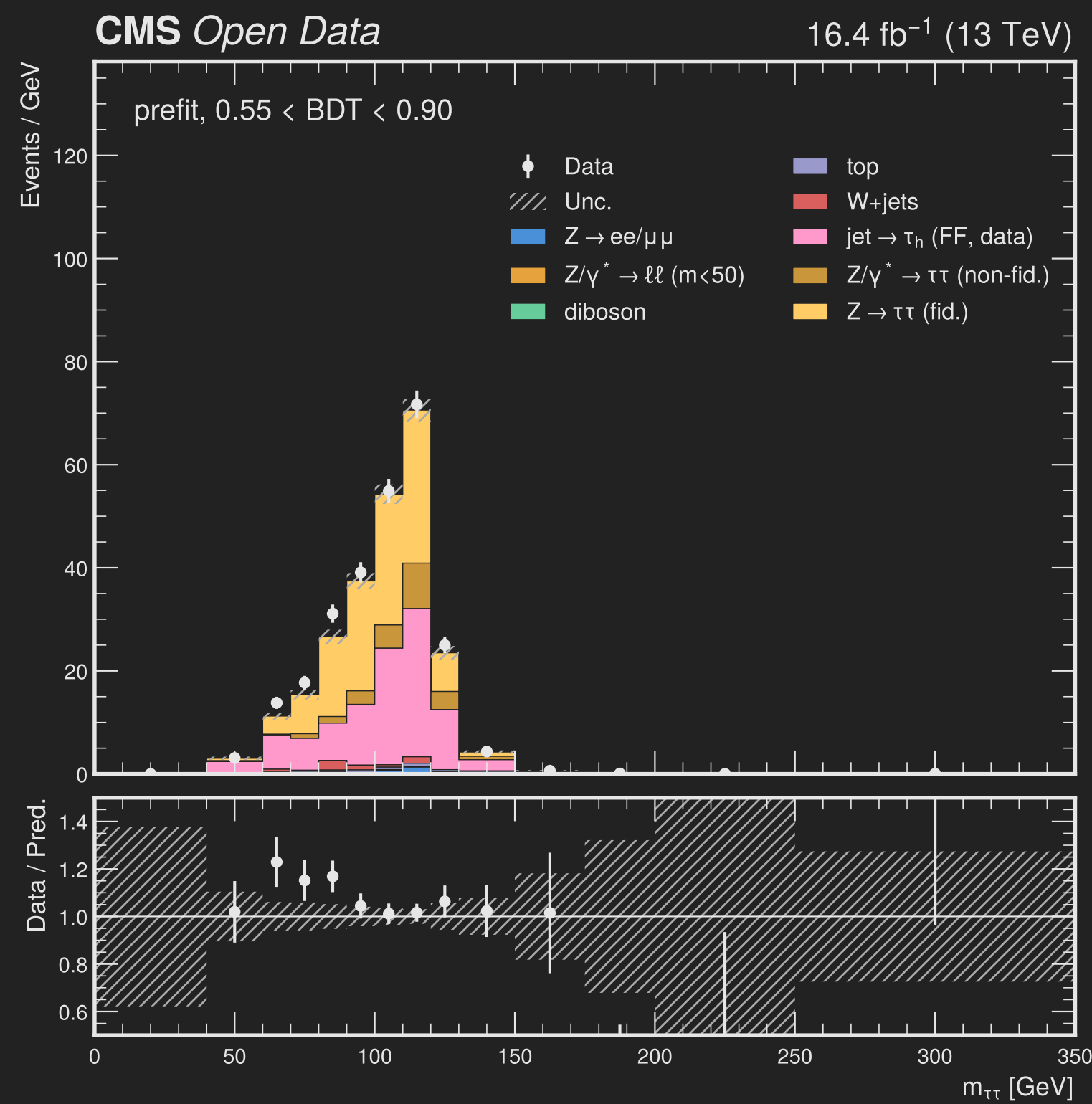
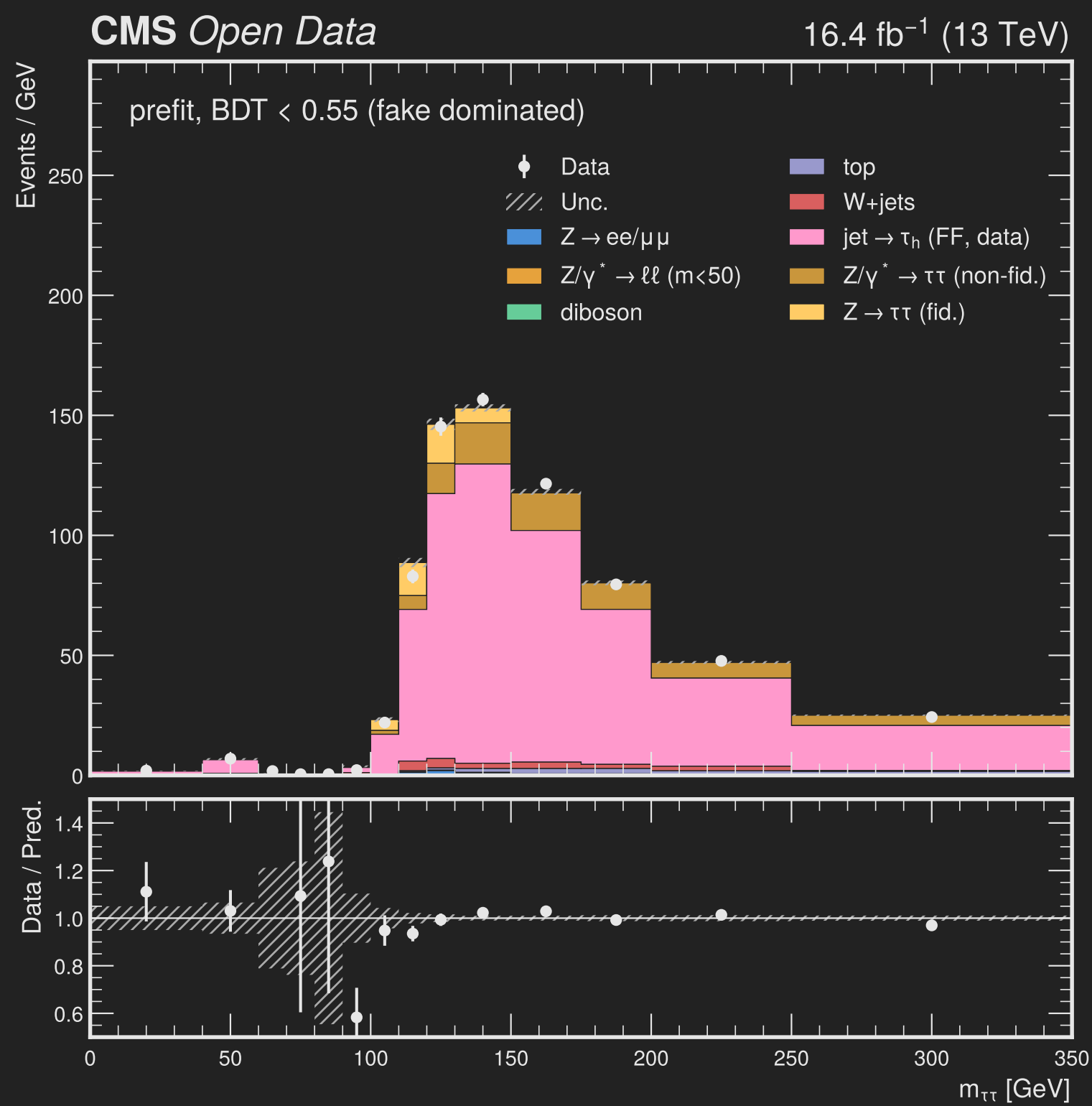
- fake NPs pulled 0.6-1.5 sigma -> the OS/SS charge correlation depends on the BDT topology: C per (era, N_{jets} , category), one NP per category
- poor goodness of fit ($p = 0.01$) -> the fake-dominated category is a sideband: fitted above 110 GeV only ($p = 0.22$)
- trigger paths and luminosity verified: both di-tau paths active and unrescaled in every certified lumisection; single-tau paths rejected (120 GeV threshold keeps 8 % of the signal, no POG scale factors)
- Tight working point run through the same chain (v2.1 cross-check), then made the nominal (v3): +11/-10 % instead of +14/-12 %, closer to NNLO

3. THE FIT: THREE BDT CATEGORIES

SR0: $BDT < 0.55$ (fake dominated; fitted above 110 GeV)

SR1: $0.55 < BDT < 0.90$

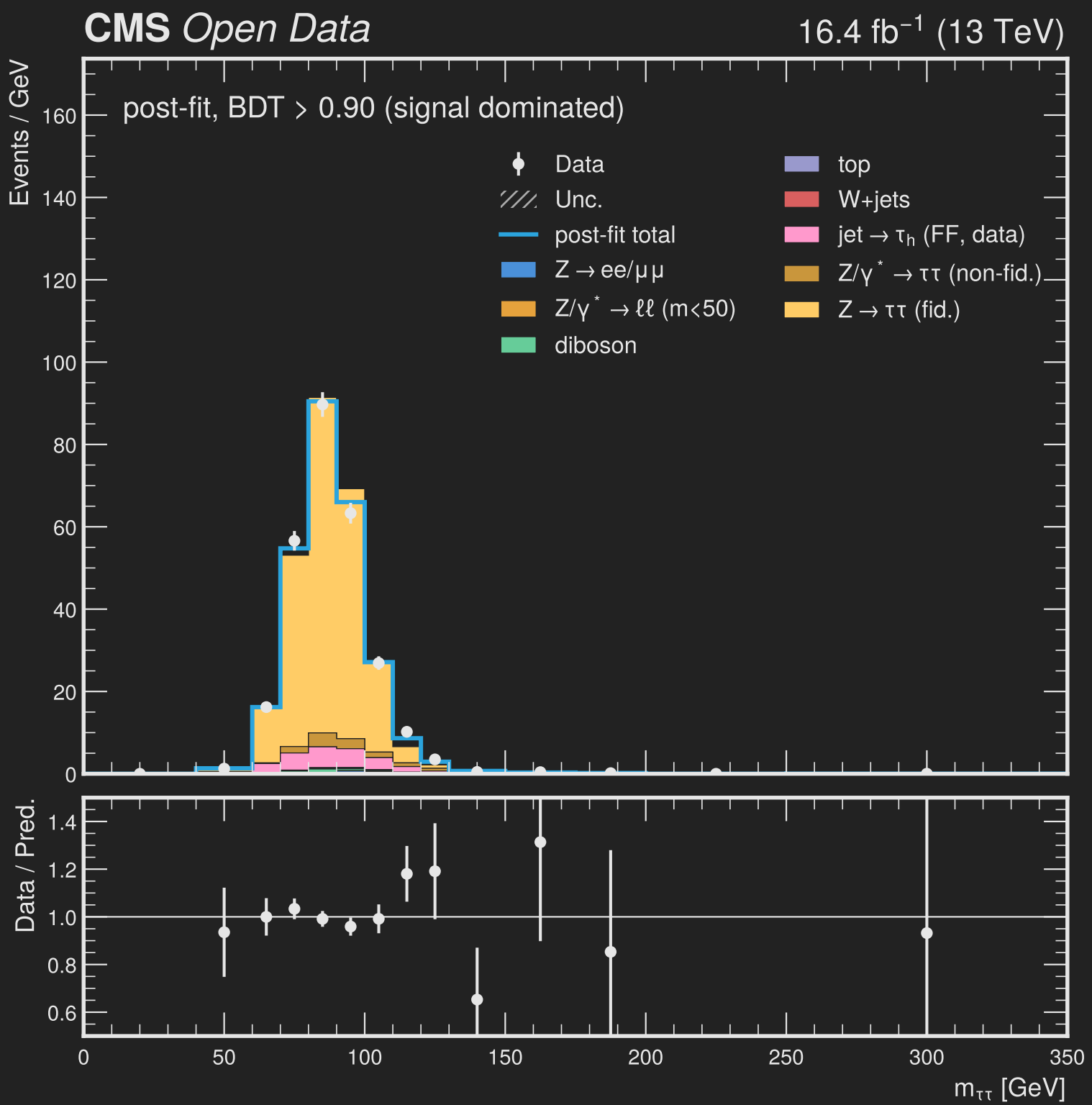
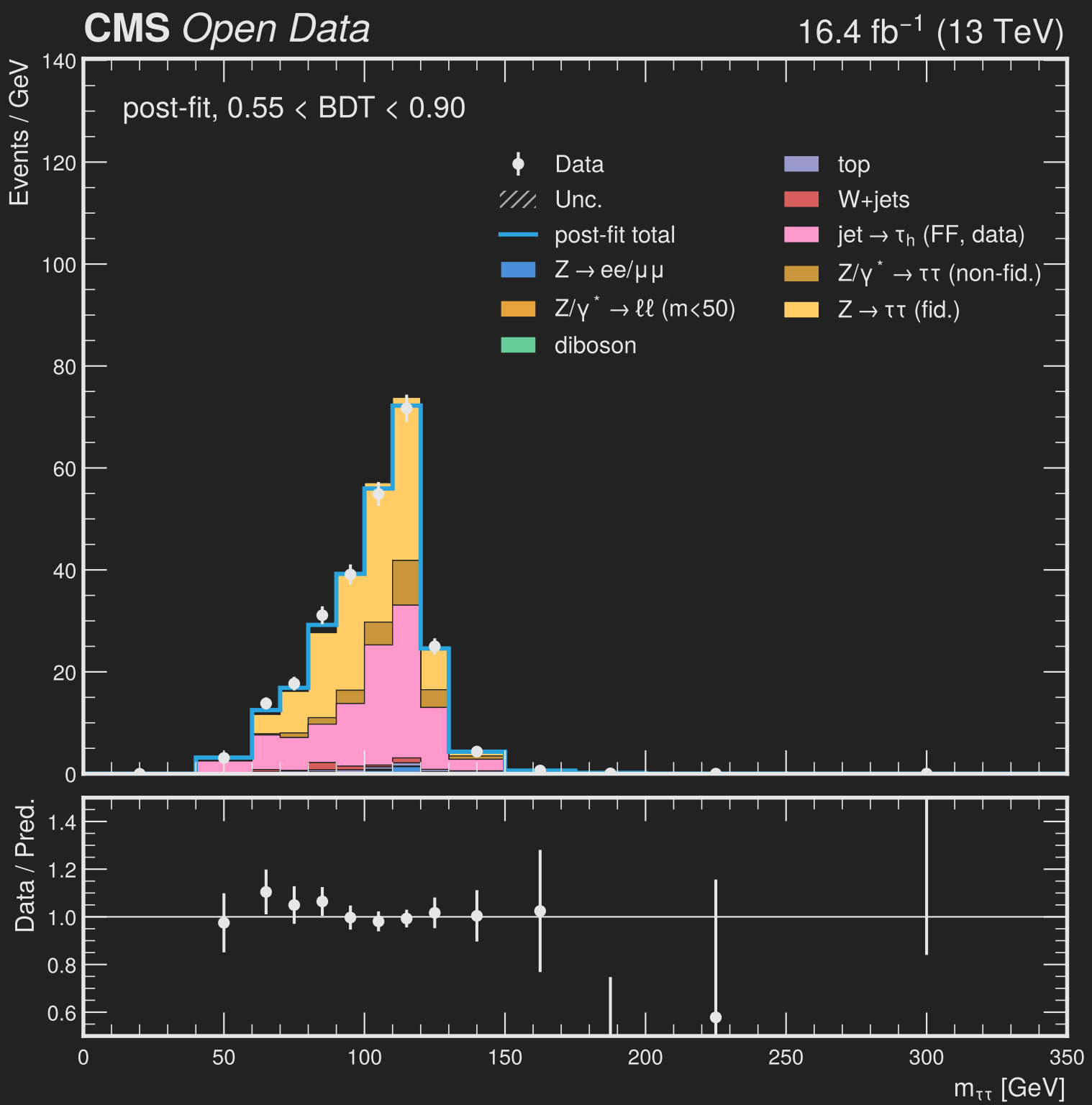
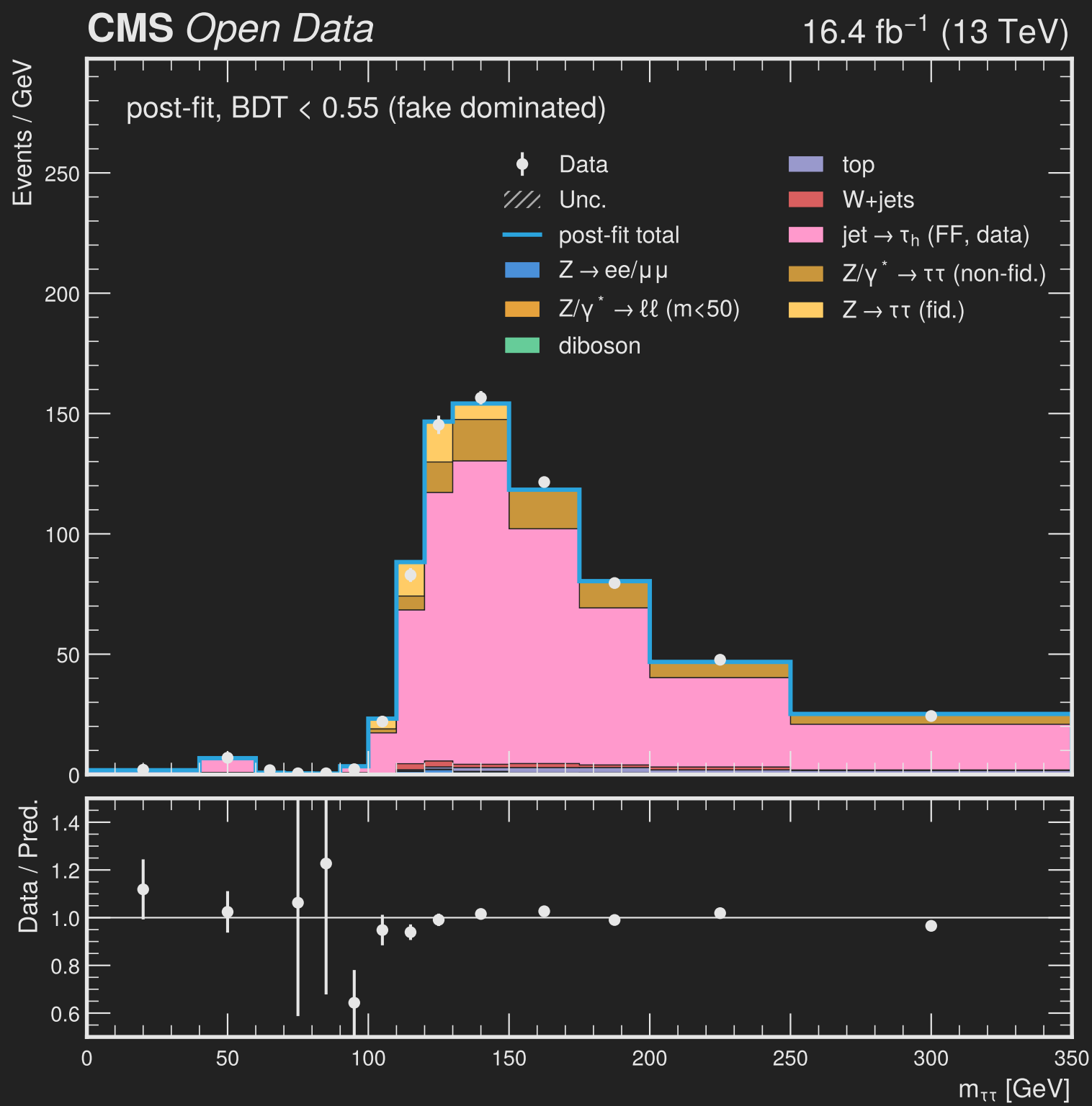
SR2: $BDT > 0.90$ (signal dominated)



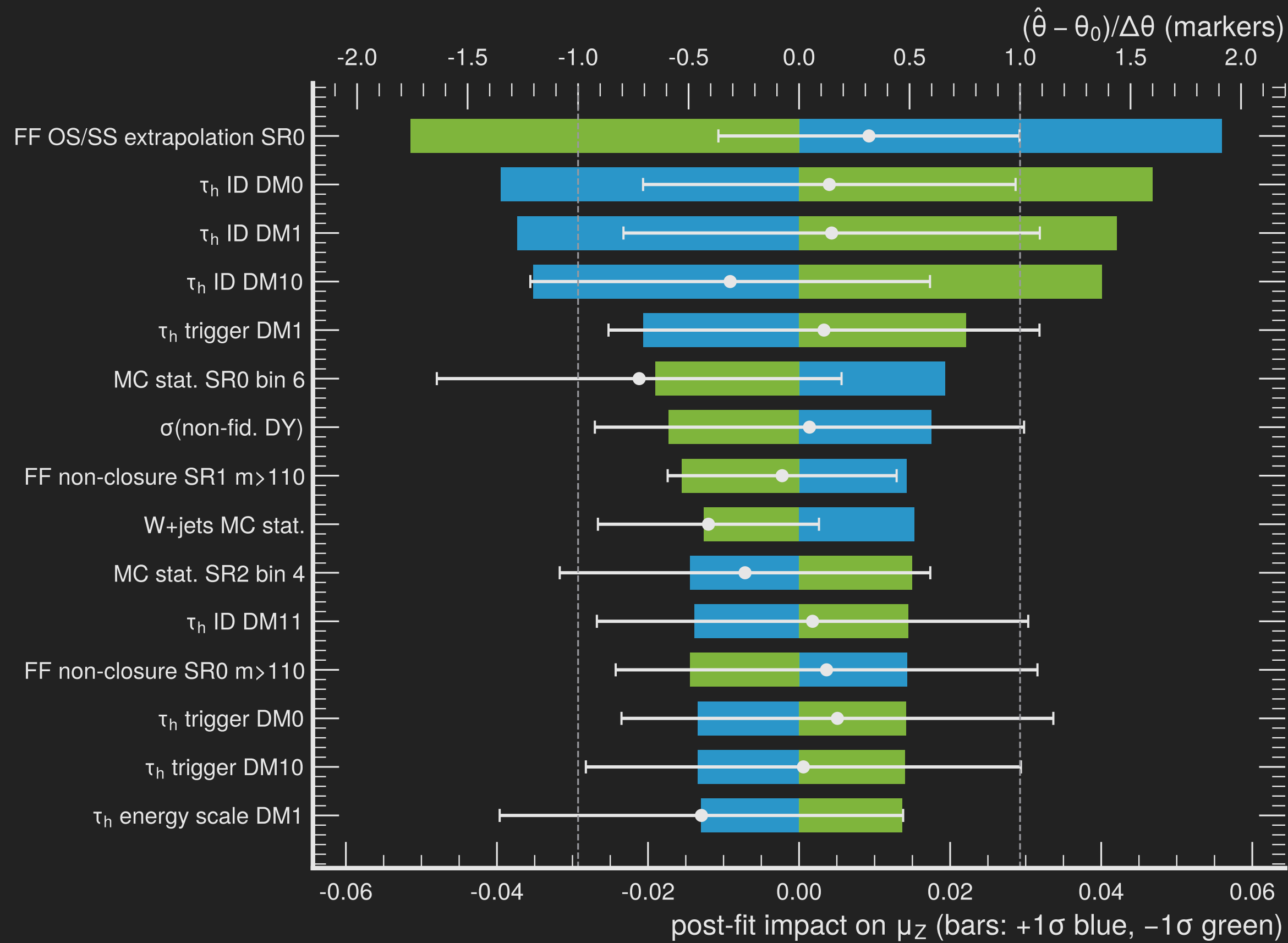
SR0 (post-fit total from TRexFitter; bins below 110 GeV not fitted)

SR1

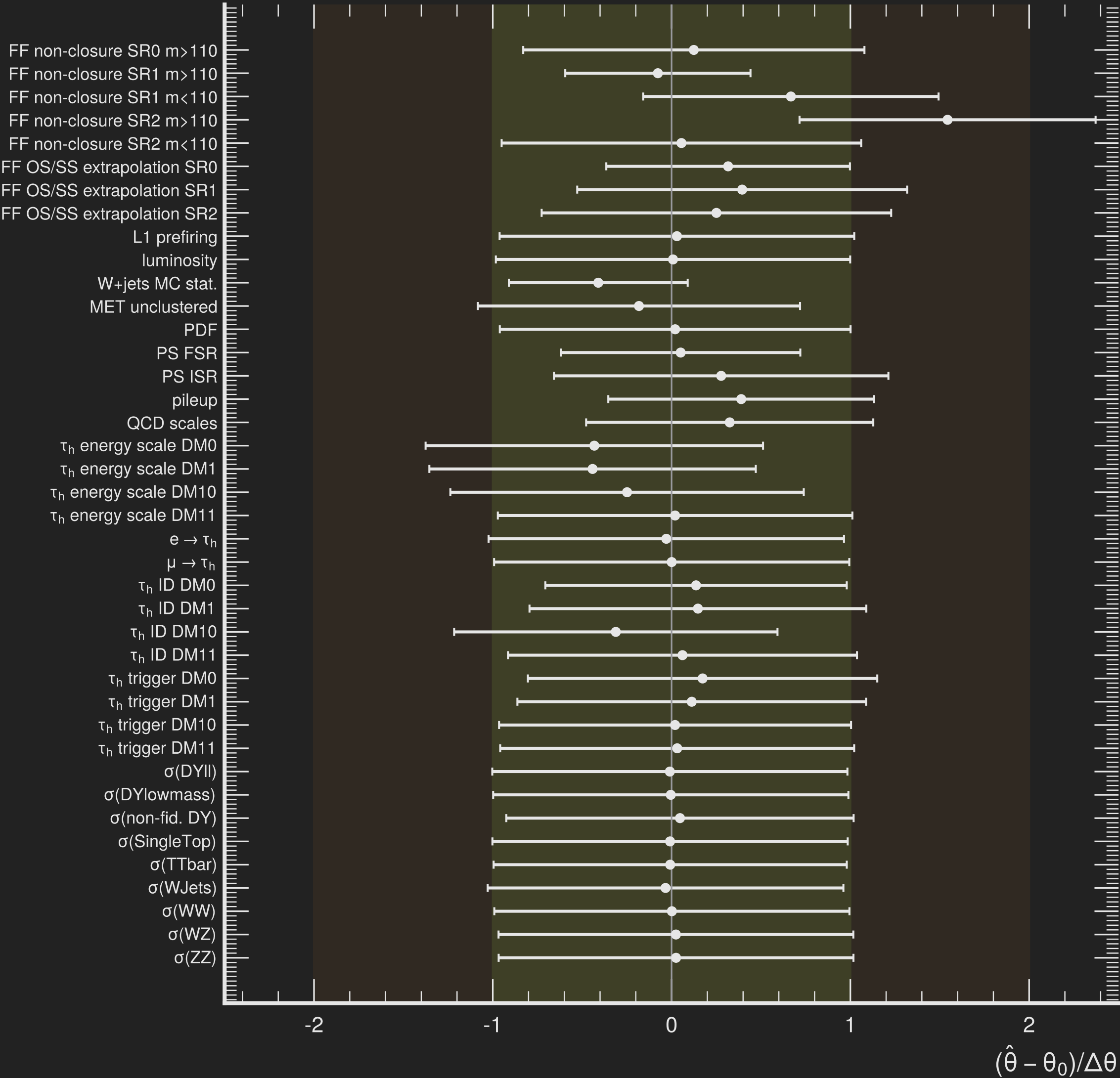
SR2: S/B = 10.9



top 15 by post-fit impact on μ_Z



all nuisance parameters



events, prefit (MC-subtracted fake factor)

sample	SR0 (< 0.55)	SR1 (0.55-0.90)	SR2 (> 0.90)	all
DYtautau	486	1131	2141	3759
DYtautau_nonfid	1972	240	124	2337
Fakes	12351	1044	197	13592
TTbar	354	41	16	410
WJets	375	62	-79	358
DYee	68	26	16	110
WW	39	2	2	43
WZ	7	7	17	30
ZZ	6	5	9	20
SingleTop	42	3	2	47
DYlowmass	16	-5	0	11
total prediction	15716	2556	2446	20718
data	15742	2704	2714	21160

4. THE K-FOLD BDT

Why

64 % fakes even at Tight (80 % at Medium); every fake uncertainty scales with B/S.
A classifier keeps the signal and sorts the events by topology.

Inputs

(mass-agnostic): $p_T(\tau_{1,2})$, ratio, $|\eta_{1,2}|$,
 ΔR , $\Delta\phi$, MET, MET significance, p_T^{vis} ,
 $\Delta\phi(MET, \tau\tau)$, $p_T(\tau\tau + MET)$, N_{jets} , jet p_T , $DM_{1,2}$
Forbidden: tau1 isolation / raw DeepTau (defines the FF regions), charge, any mass.

Training

signal = fiducial Z->tautau MC (29,117 events, inclusive + jet-binned);
background = the fake estimate itself (AR data x FF, 120,805 events).
XGBoost, depth 4, 300 trees, both classes with equal total weight.

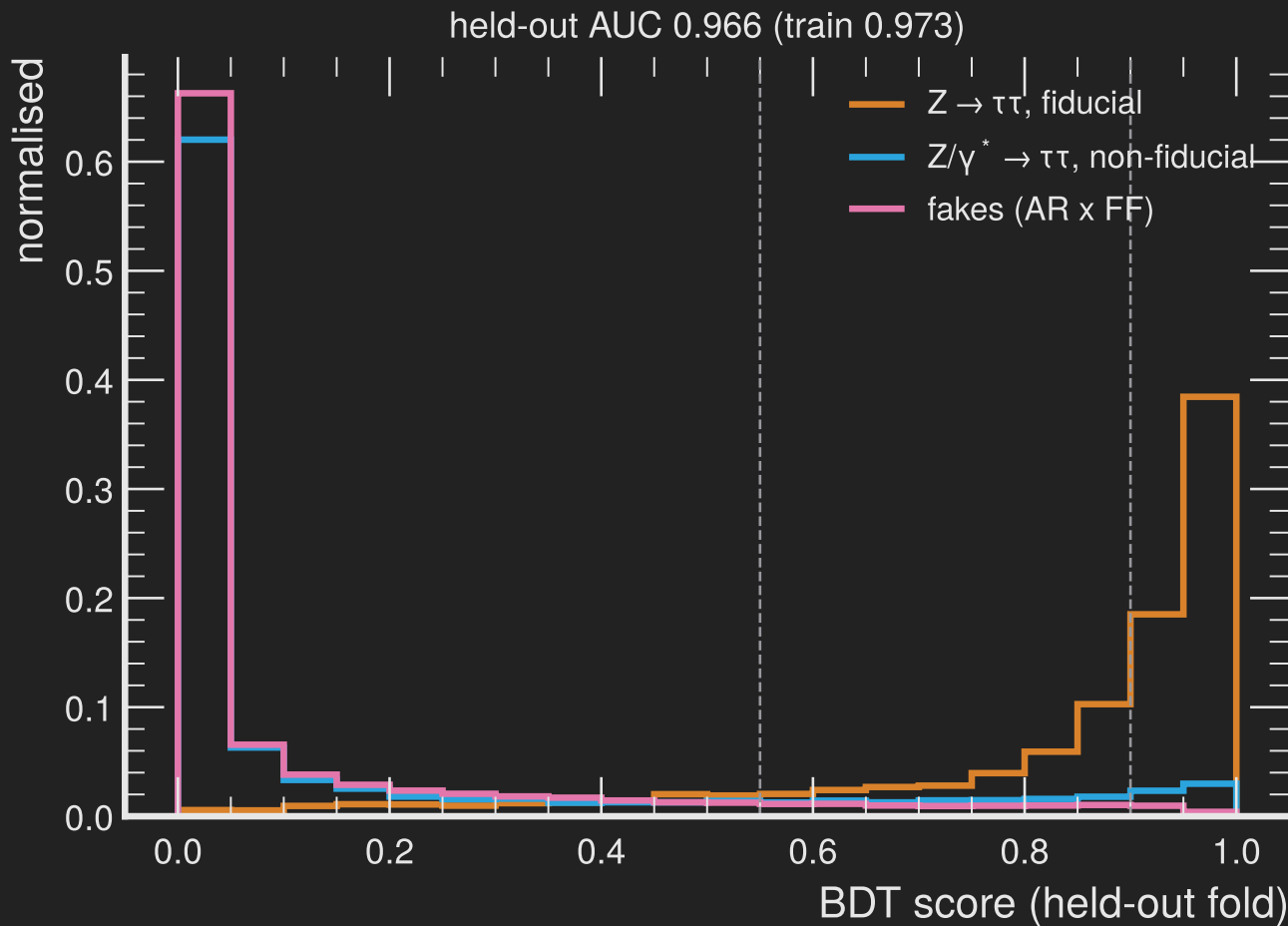
5 folds by event number

AR data, same-sign data, SR data, every
MC sample and every kinematic variation are scored by the model
that never saw them. Held-out AUC 0.966 (train 0.973).

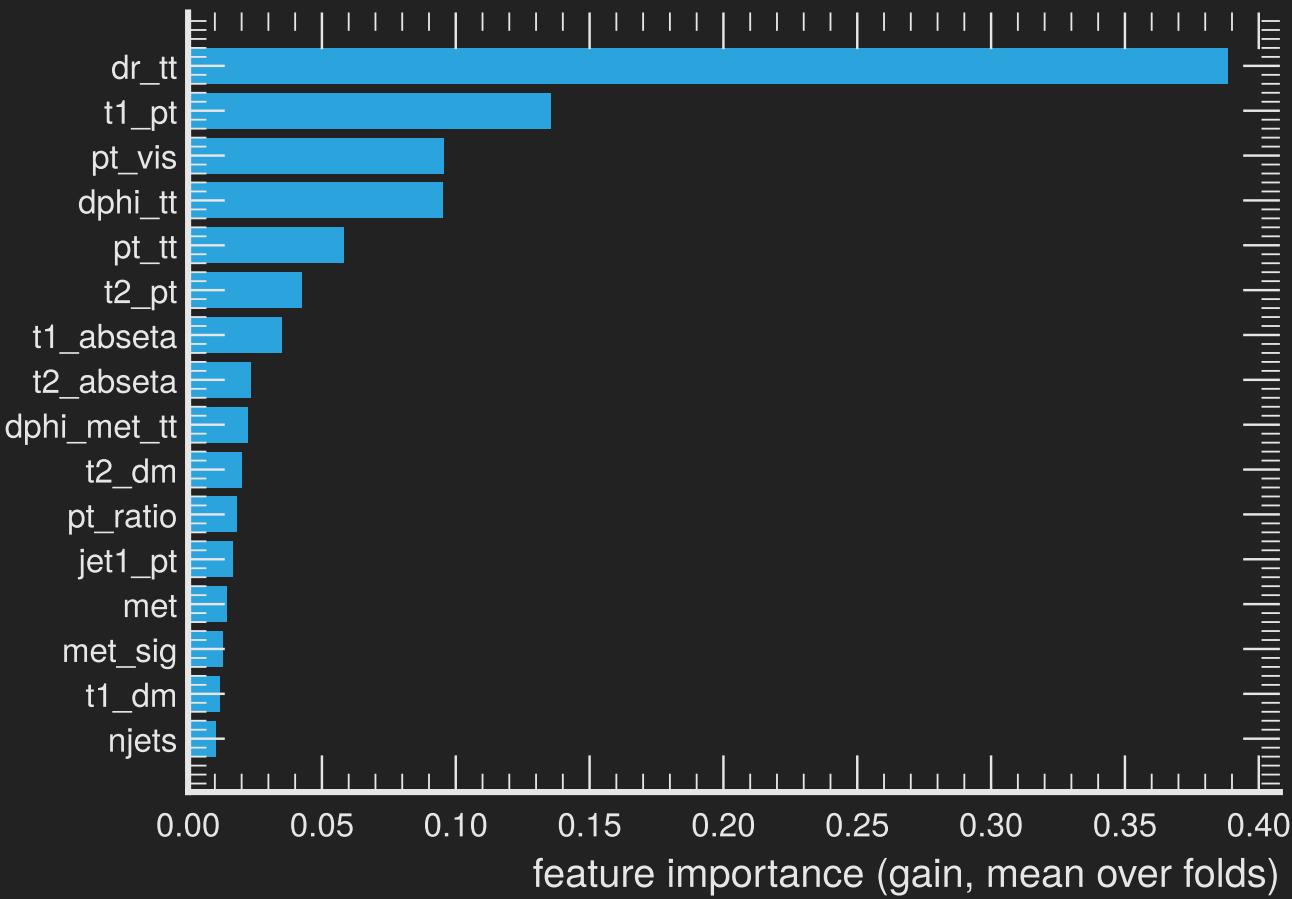
Categories

< 0.55 (fake dominated), 0.55-0.90, > 0.90 (S/B 10.9).
Stat-only sensitivity 2.01 % -> 1.93 % (fakes fixed).

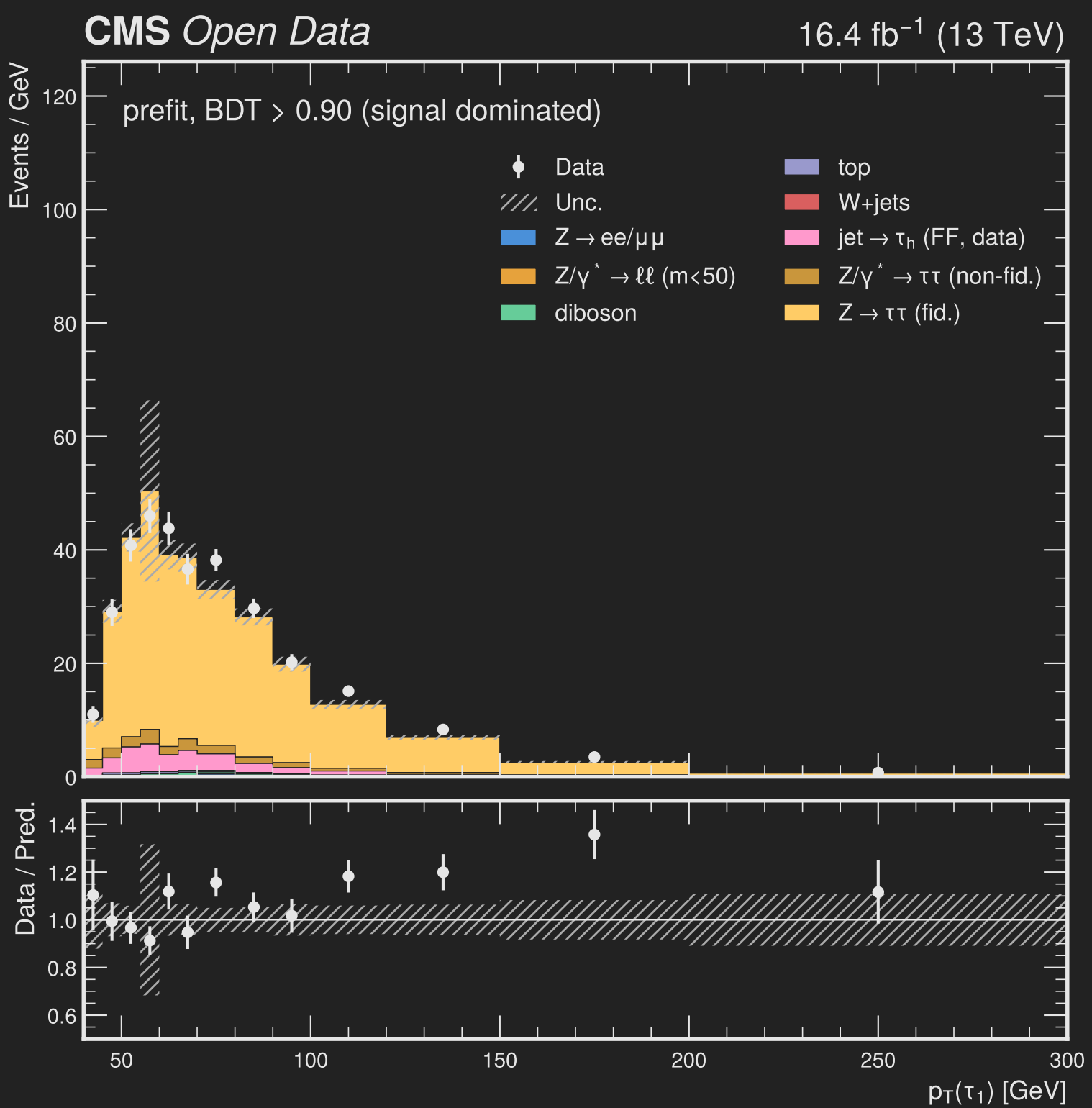
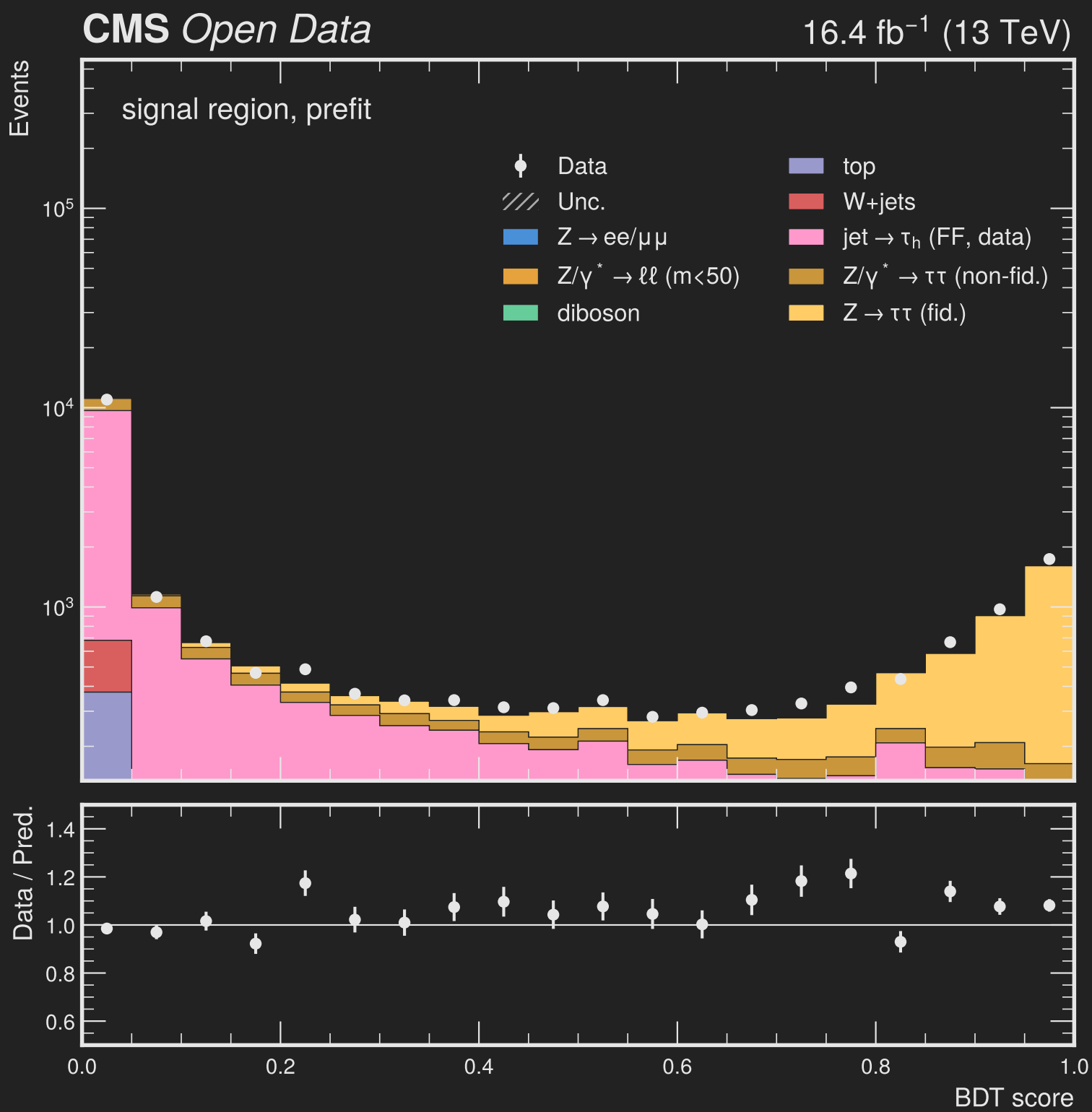
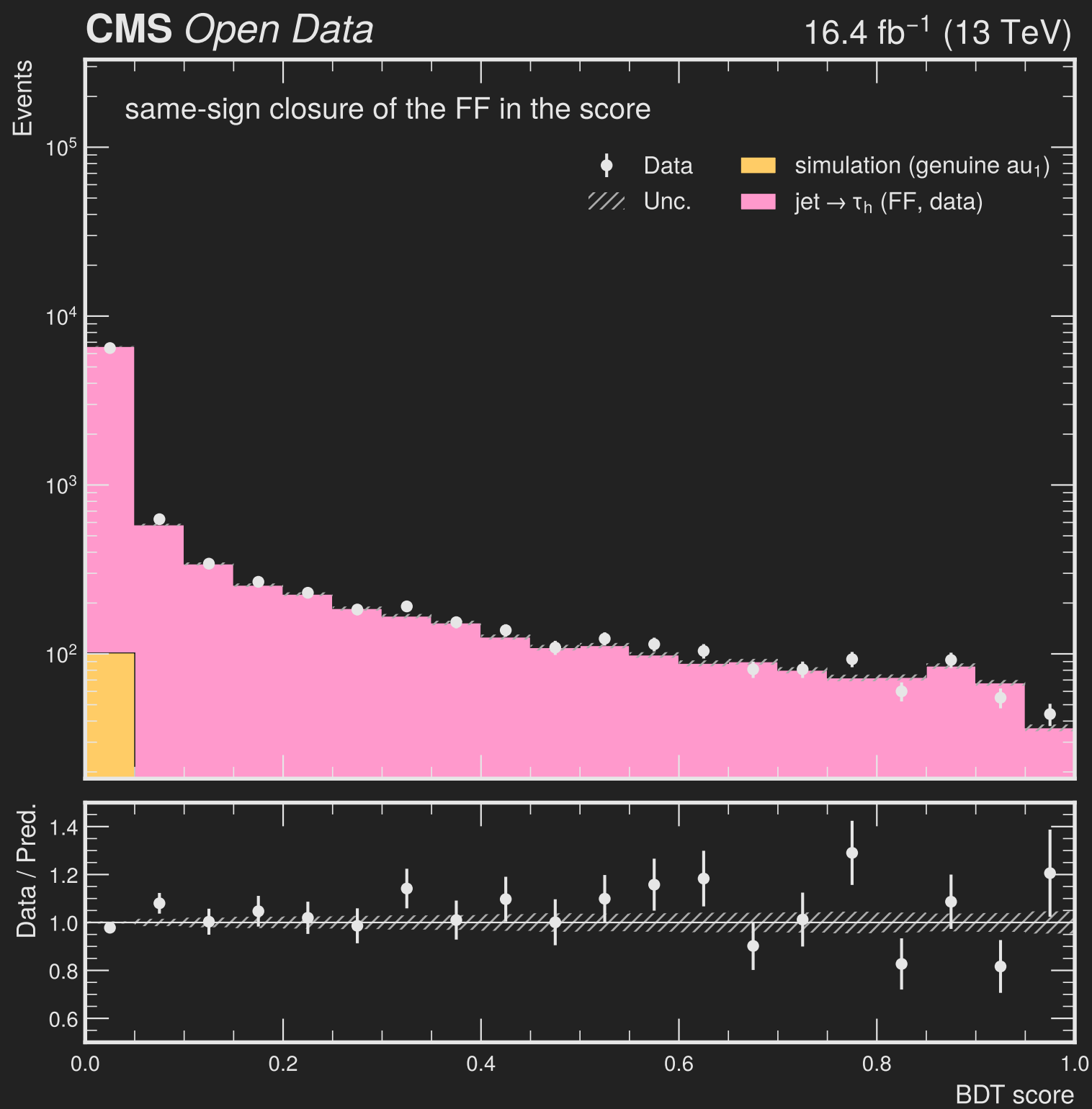
score shapes (held-out folds)



feature importance



same-sign closure in the score: SS_T vs $FF \times SS_L$ (MC subtracted) region, prefit: the fake-dominated bins close, the excess is ~ 2
SR2: $p_T(\tau_1)$ data vs prediction (the trend behind section 7)



$FF = [N(SS, \tau_1 M, \tau_2 M) - MC] / [N(SS, \tau_1 VVVL\Lambda \rightarrow M, \tau_2 M) - MC]$
 $\times f(|\eta_1|) g(p_{T,2})$; applied to OS events with τ_1 failing Medium,
times $C_{OS/SS}(era, N_{jets})$ from the τ_2 -anti-isolated sideband.

era x DM x N_jets x pT = 120 bins, ~10 % stat. each (per event into the template variance).

MC subtraction

genuine tau1 = 1-2 % of the AR but ~6 % of the signal,
~4 % of the C numerator; C inclusive 1.056 (per category in the fit).

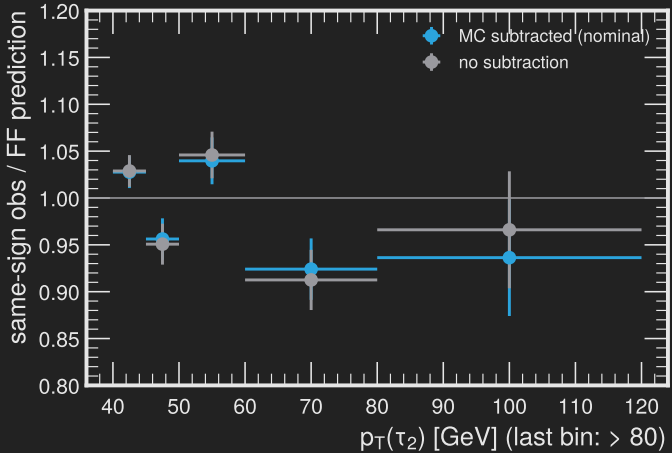
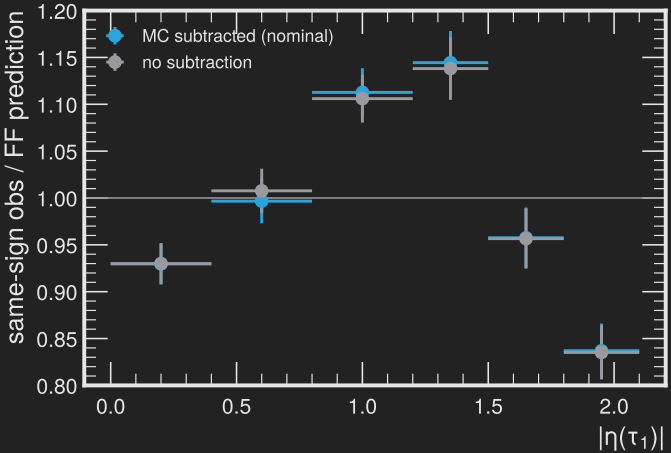
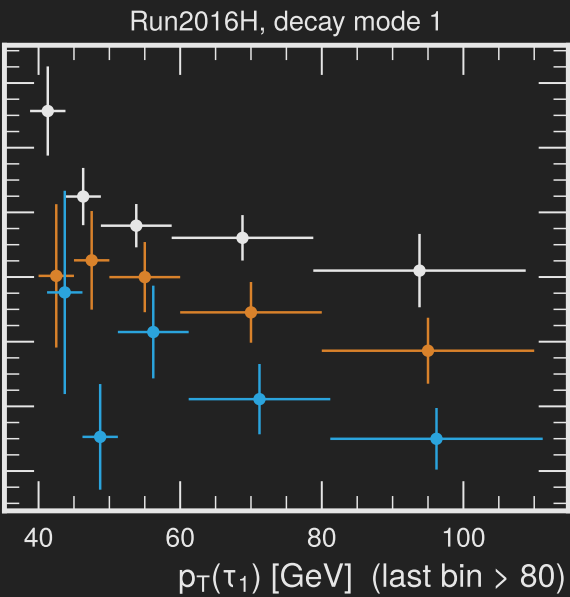
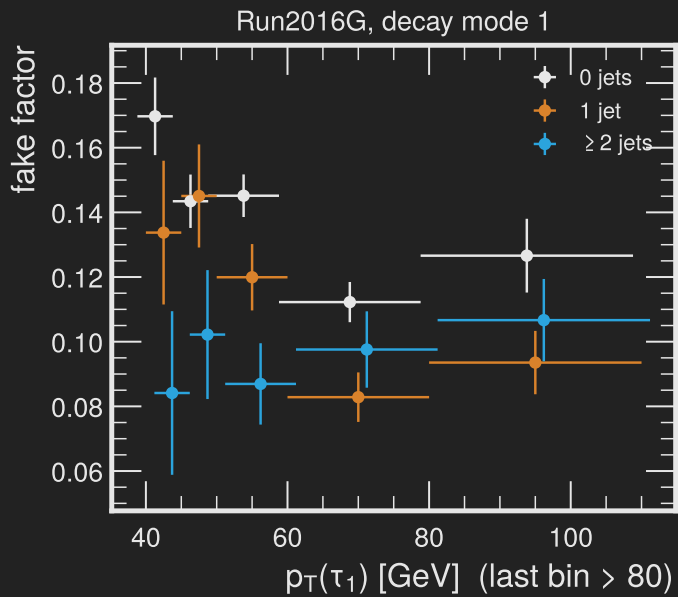
W+jets subtracted with uniform weights (raw weights 20-200, one at -63).

Closure corrections

(derived at Medium in v2, re-derived at Tight): the FF varies by +/-15 % with $|\eta(\tau_1)|$
(same shape
in both eras, all pT, N_jets and DMs: DeepTau's jet rejection is not eta-flat)
and by -6 % with $p_T(\tau_2)$. Residual non-closure per category and mass
region -> one nuisance parameter each: c0/lo 14 %, c0/hi 1 %, c1/lo 7 %, c1/hi 21 %, c2/lo 15 %, c2/hi 44 %.

fake factors, decay mode 1

closure corrections

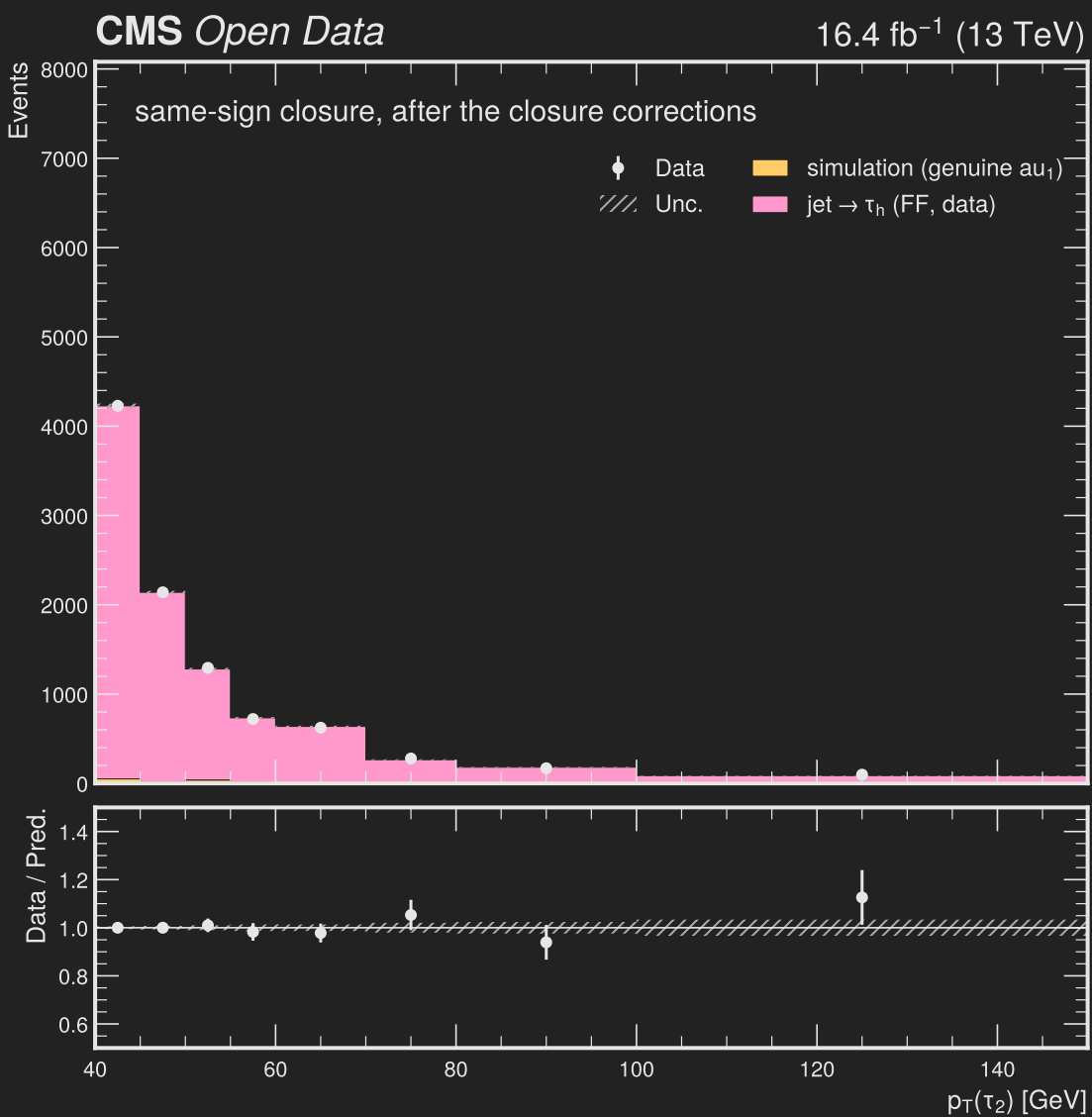
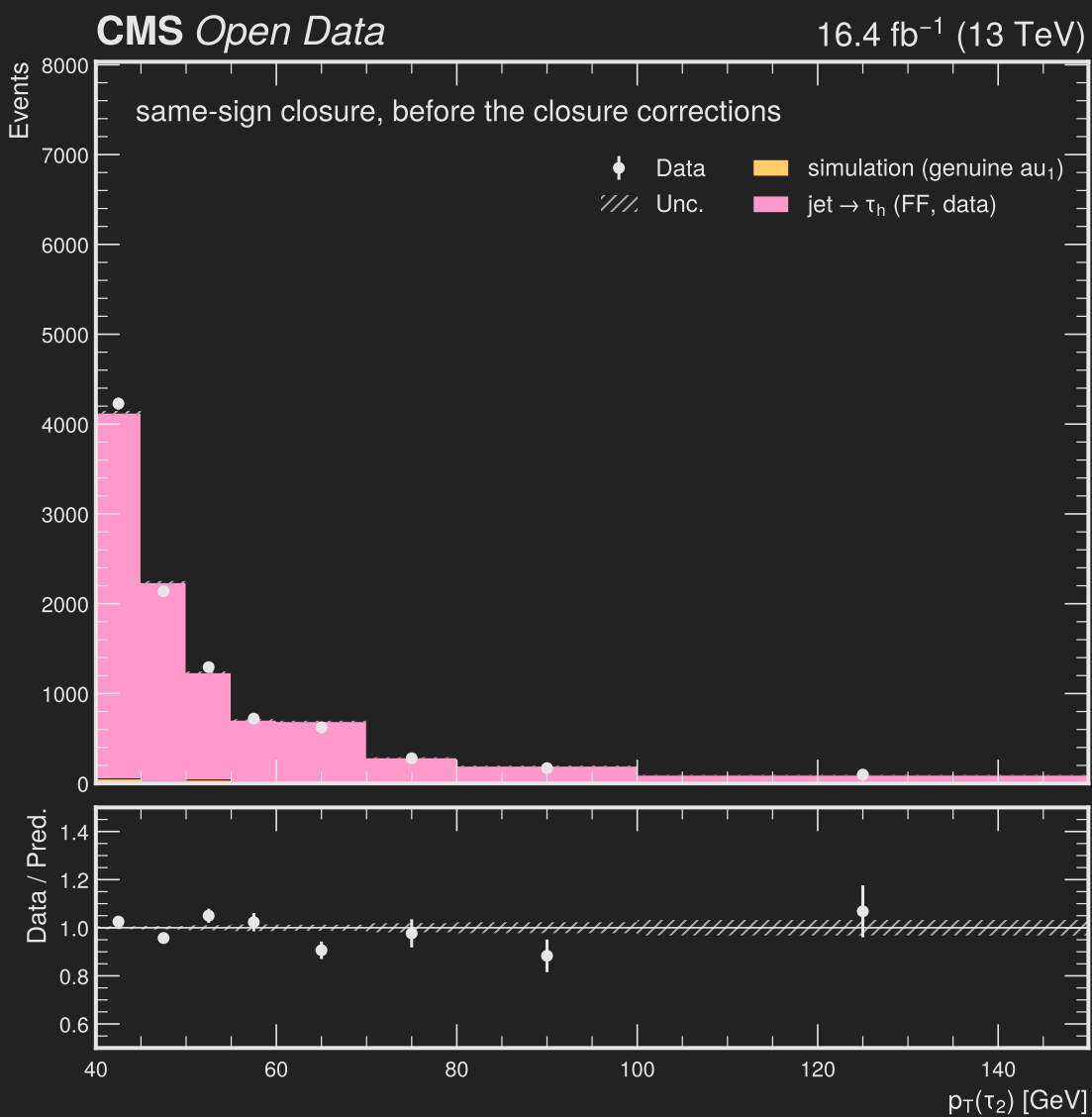
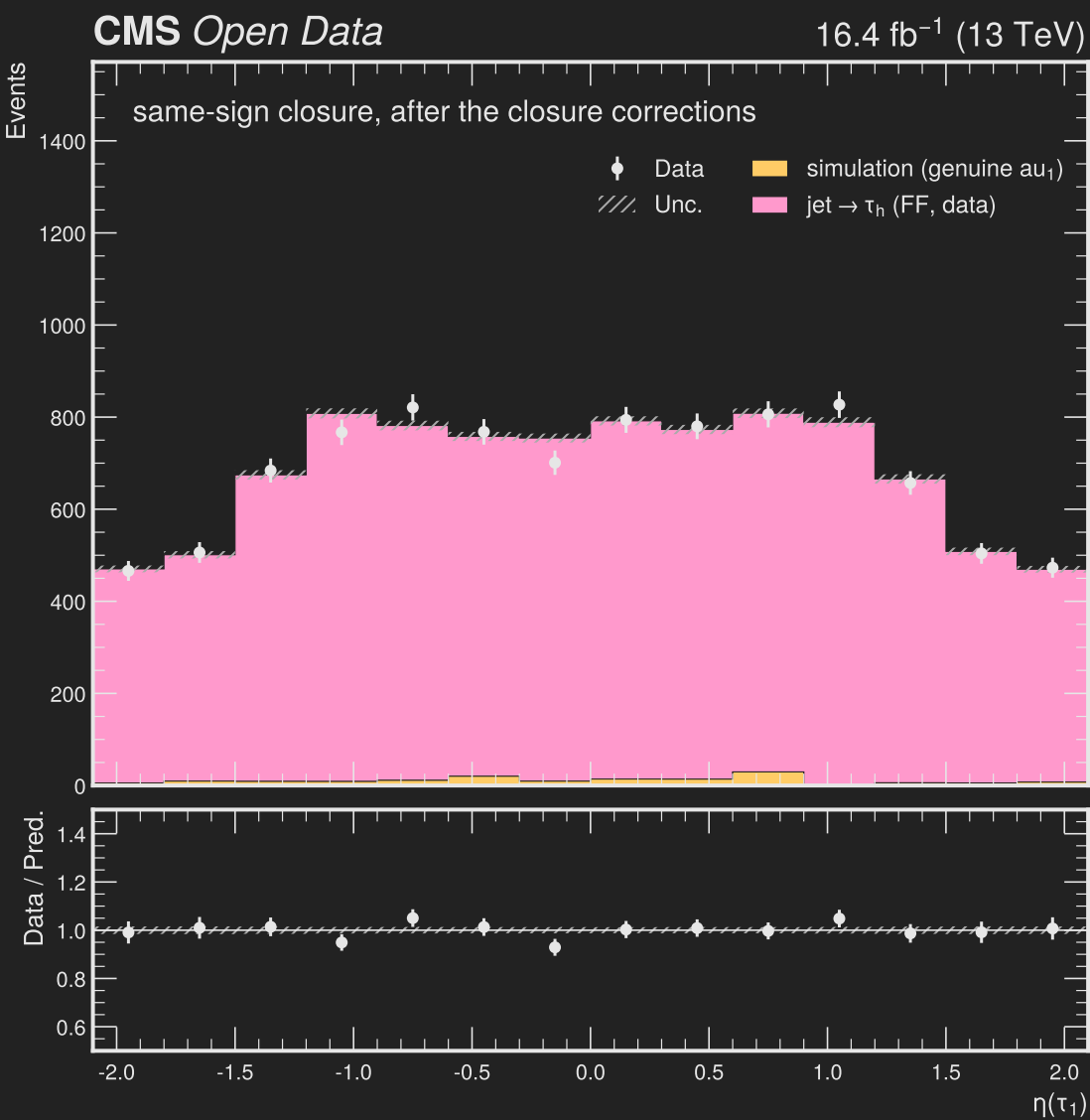
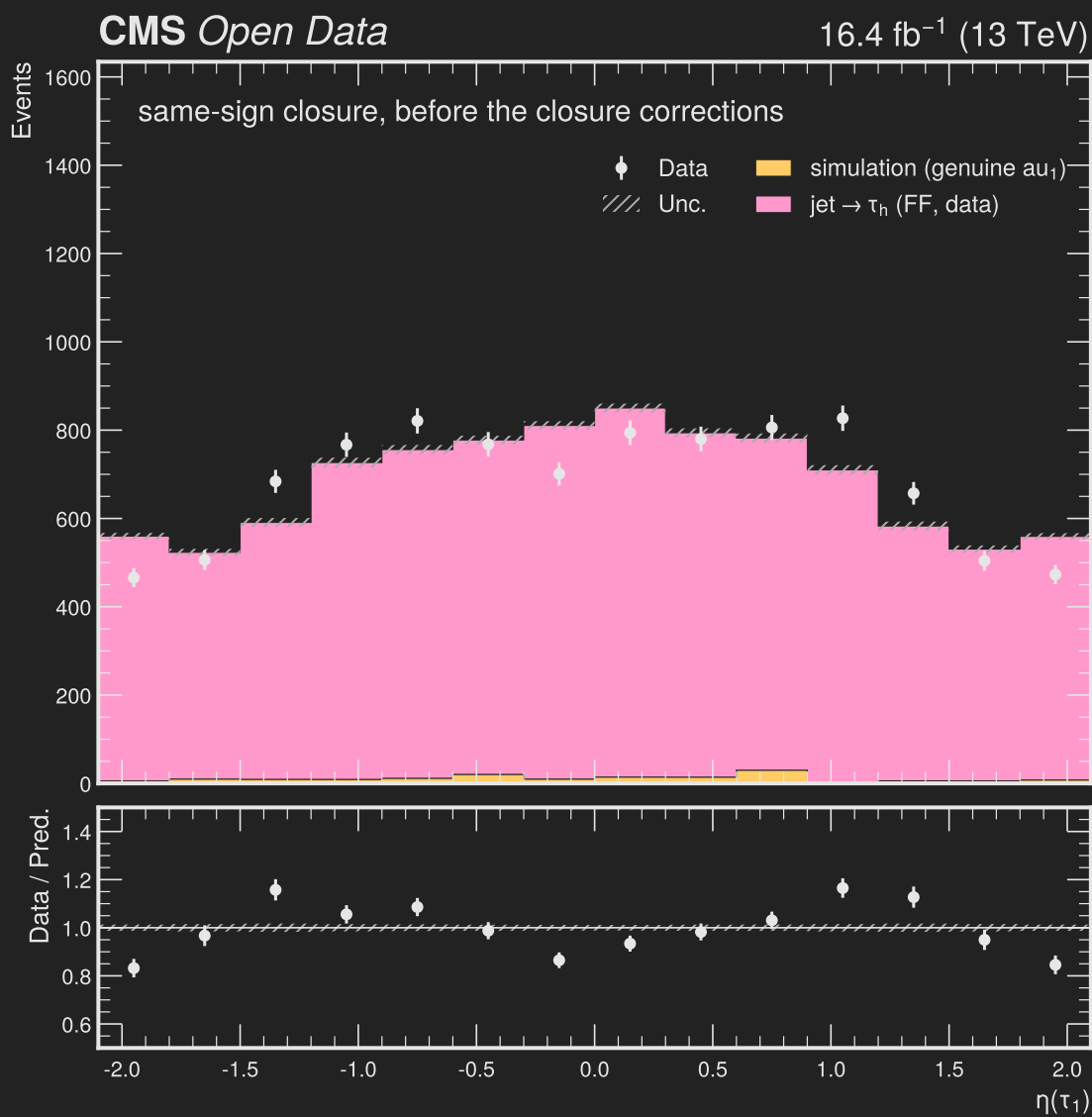


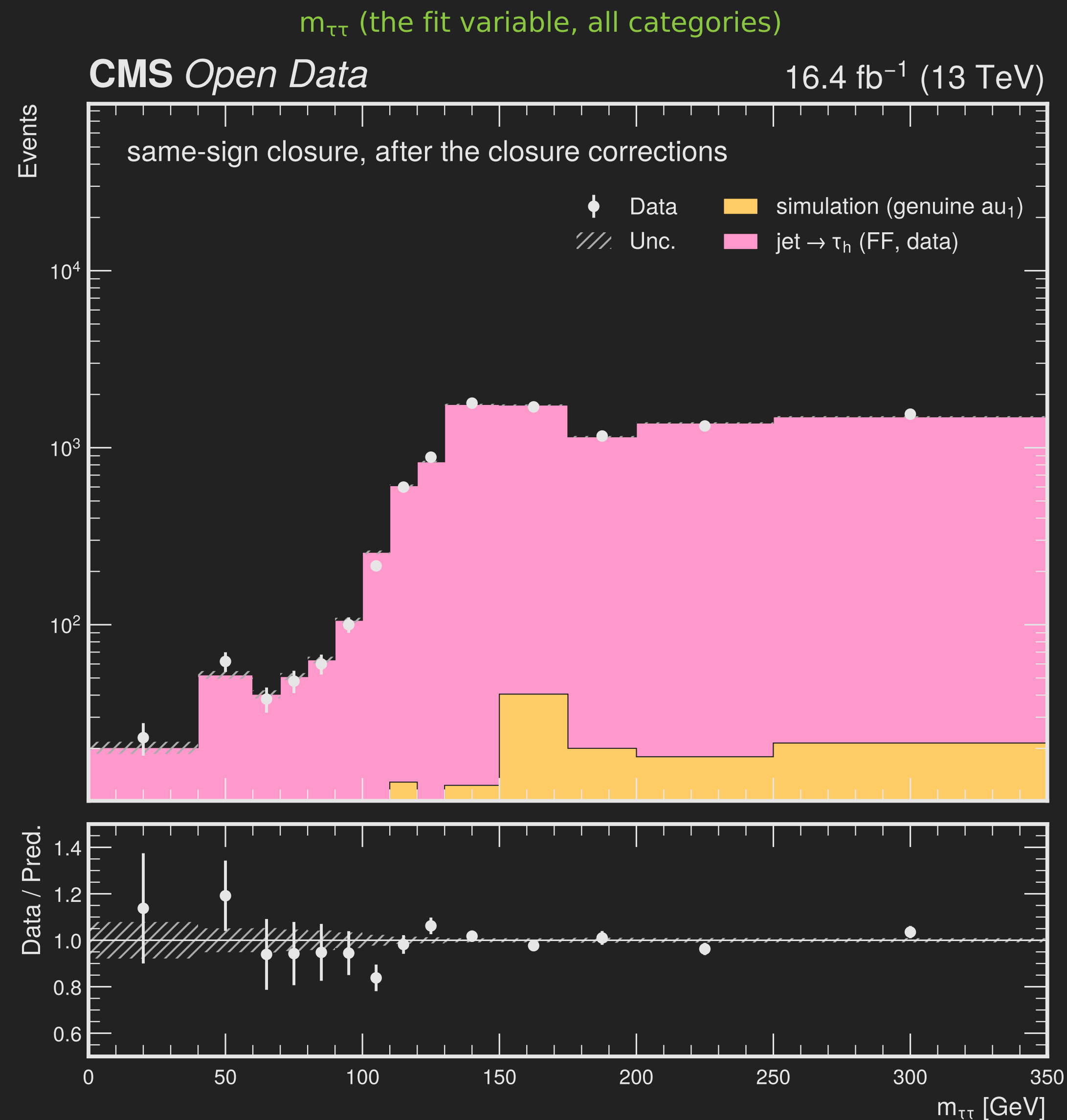
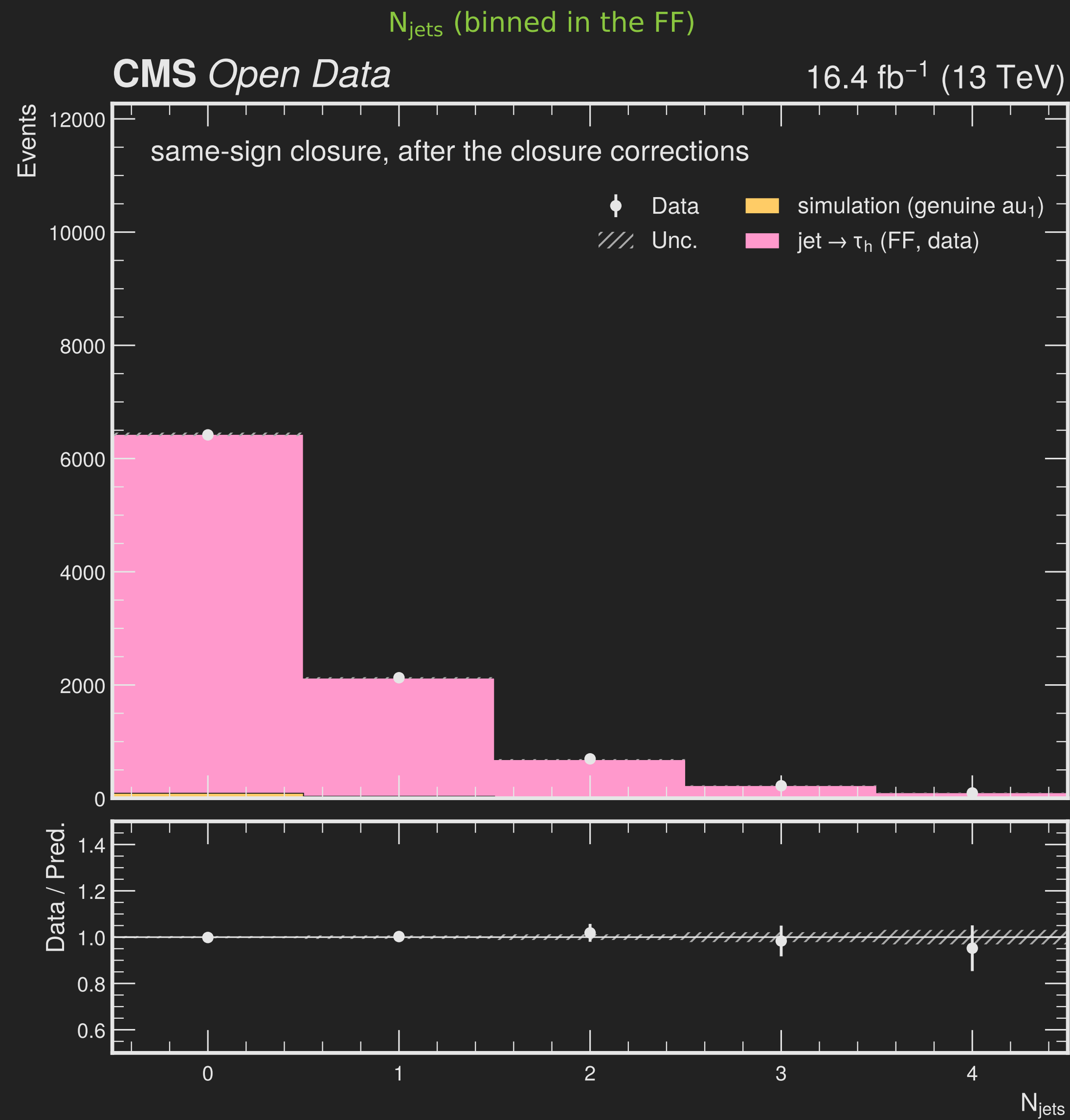
$\eta(\tau_1)$ before

$\eta(\tau_1)$ after

$p_T(\tau_2)$ before

$p_T(\tau_2)$ after





6. SIGNAL DEFINITION, MASS, SELECTION

Fiducial volume

60 < m_LHE < 120 GeV, both taus hadronic, both visible taus
pT > 40 GeV, |eta| < 2.1. A = 0.231 % (+-3.7 %), sigma_fid(pred) = 4.50 pb.

Of the selected Z/gamma* -> tautau

62 % fiducial (-> DYtautau, x mu_Z),
30 % has m_LHE > 120 GeV, 6 % migrates across the 40 GeV / |eta| cuts, 1.5 % has a tau->e/mu leg.

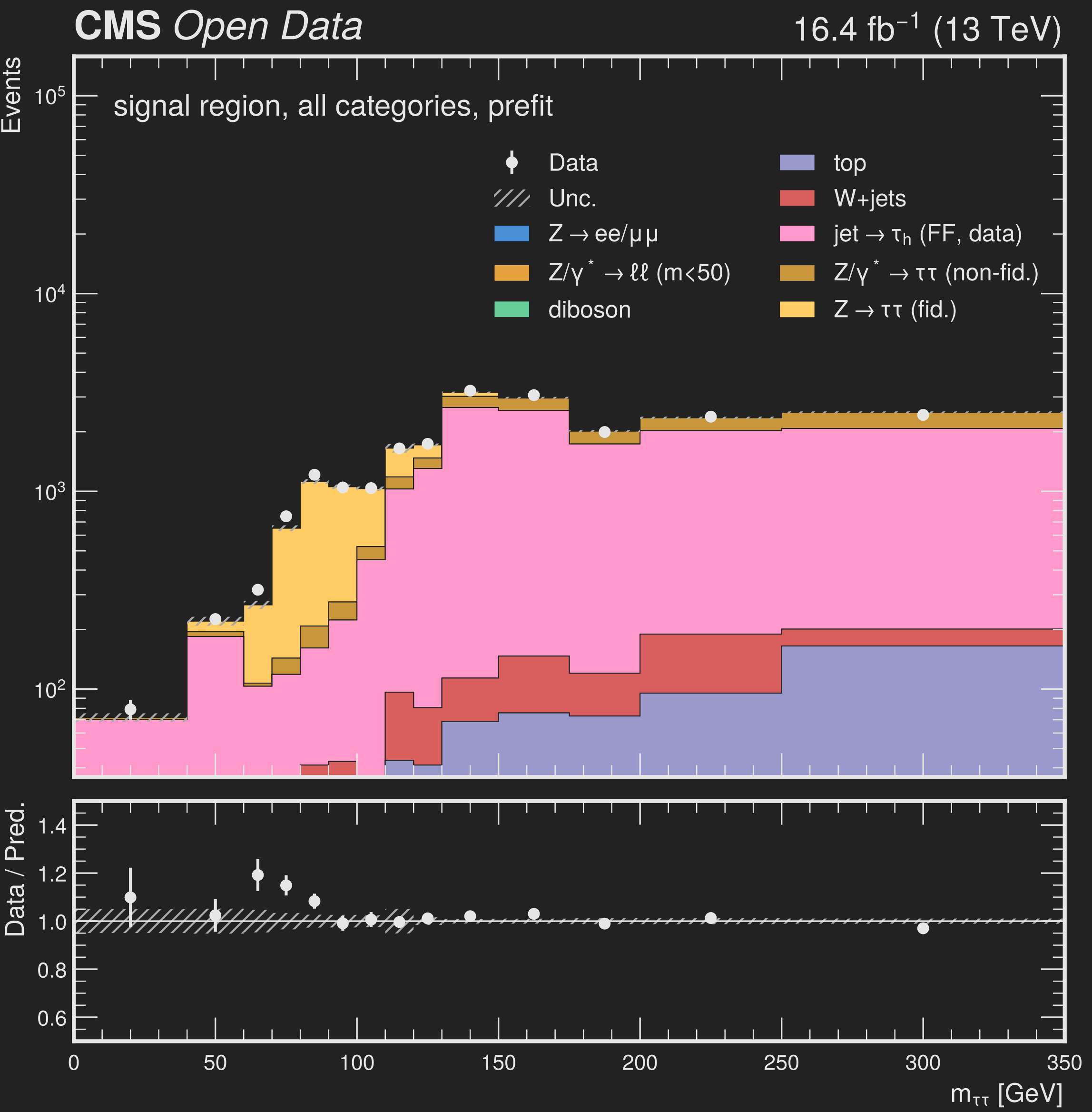
The two 40 GeV cuts enrich the gamma* continuum by two orders of magnitude: 88 % of the Z->tautau above m = 130 GeV (the fake sideband) has m_LHE > 120 GeV.

v2

DYtautau_nonfid is a background (5 % normalisation NP + the scale/PDF/PS variations acting on the non-fid/fid ratio, all experimental NPs correlated).
sigma_fid = mu x sigma_fid(pred) is literally true; the sideband no longer contains mu-dependent signal (REVIEW.md 3.1).

Generator cross-check

fiducial C(LO madgraph)/C(NLO) = 0.841 -- the LO sample's softer visible-tau spectrum meets the trigger turn-on. Reported, not an NP (docs/07).



≥ 4 neutrinos escape: m_{vis} peaks at $0.8\ m_Z$.

Collinear approximation

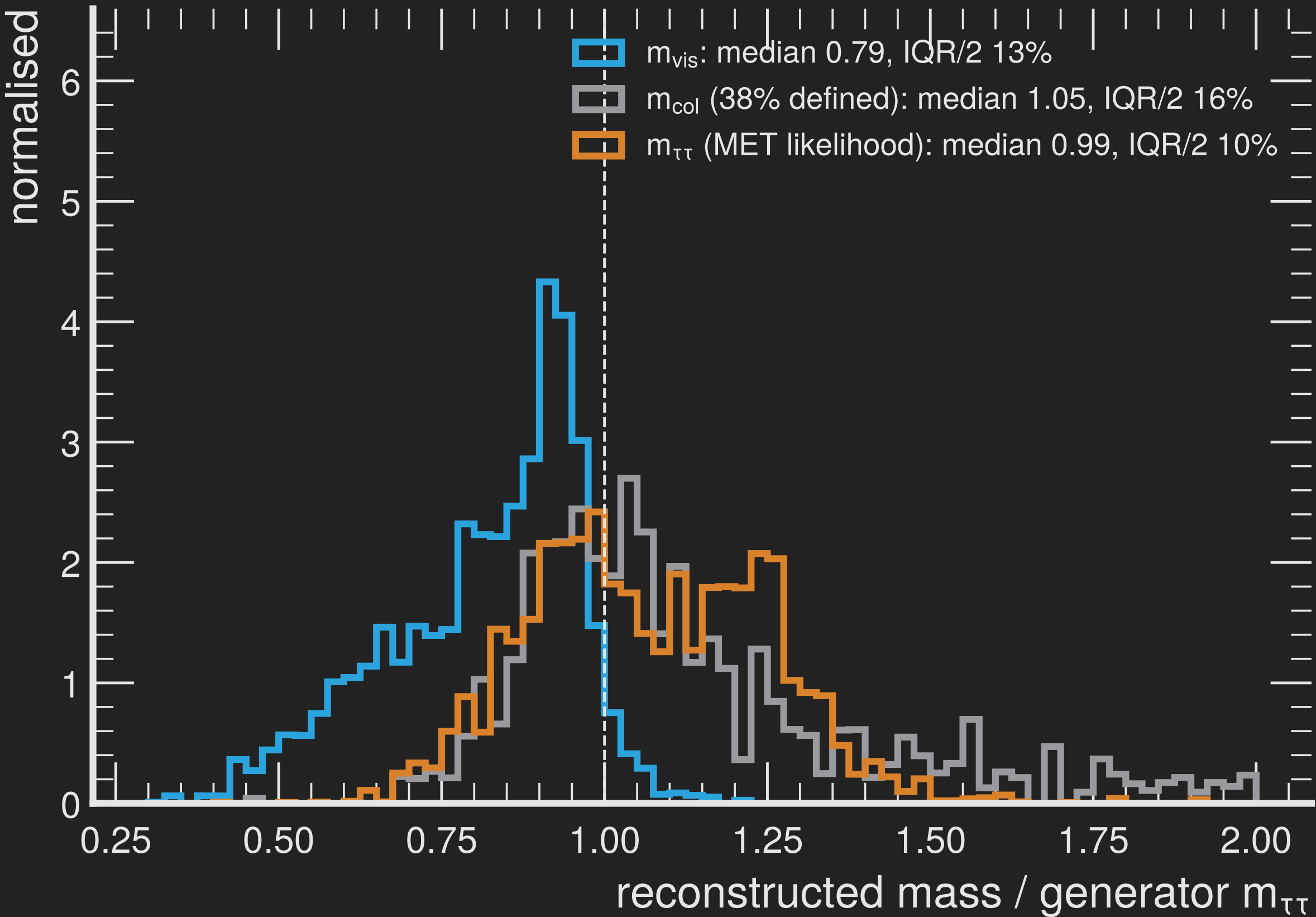
singular for back-to-back taus, defined in 38 %.

Likelihood mass

(FastMTT / SVfit-lite idea): scan the visible fractions
(x_1, x_2), weight = MET transfer function (per-event covariance)
 $\times$ flat two-body phase space $\times 1/m^2$; posterior median.

scale 0.99, resolution 11 % (m_{vis} : 0.80, 13 %); always defined.

Separation from fakes about equal to m_{vis} ; $m_{\tau\tau}$ is the physical mass
and gives a clean high-mass sideband for the fake normalisation.



selection

object	requirement
trigger	DoubleMedium(Combined)IsoPFTau35_Trk1_eta2p1_Reg (G: Iso, H: CombinedIso); both legs matched to HLT tau objects
tau_h	pT > 40 GeV, eta < 2.1, dz < 0.2 cm, DM 0/1/10/11; DeepTau VSe VVLoose, VSmu VLoose; VSjet Tight (tight) / VVVLoose & !Tight (loose)
pair	most isolated pair, dR > 0.5; tau1 = leading pT (the fake-factor leg)
veto	no extra e / mu: orthogonal to e tau_h, mu tau_h, ee, mumu
simulation	events whose leading tau is a jet are dropped (covered by the fake factor)

cutflow

step	Run2016G	Run2016H
NanoAOD	79.6 M	76.8 M
skim (certified, trigger, >= 2 candidates)	2.81 M	2.22 M
pair + MET filters	2.03 M	1.62 M
trigger match	1.99 M	1.59 M
lepton veto	1.99 M	1.59 M
signal region (OS, both Tight)	-	-

7. FINDINGS AND NEXT STEPS

1. $\sigma(60-120) = 2082^{+222}_{-194} \pm 76$ pb, $\mu_Z = 1.07$: 0.7σ above NNLO and $Z \rightarrow \mu\mu$ ($\mu_Z = 0.99$). Total $^{+11}_{-10}$ % (v2.1 Medium: +14/-12 %, v1: +19/-16 %), data statistics 2.1 %.
2. The limit is external: the TauPOG tau ID scale factors (8 %); Z->tautau is how those SFs are measured, and the two POG prescriptions (DM- vs pT-binned) differ by 14 % on the yield.
3. In the signal-dominated category data/prediction rises with the visible tau pT (1.0 at threshold, 1.2 above 70 GeV): part of $\mu > 1$ comes from there (the fit quality itself is fine, $p = 0.25$, once the fake sideband is restricted to $m > 110$ GeV). Trigger turn-on modelling or the NLO Z pT spectrum; an in-situ trigger measurement is the next step.
4. The fake factor closes only with N_jets, era and $|\eta(\tau_1)| / pT(\tau_2)$ corrections; the FF is eta-dependent by +/-15 % because DeepTau's jet rejection is not. The MC subtraction is not optional (bias -0.05 on mu; unsubtracted fakes 7 % high where there is no signal).
5. 38 % of the selected Z->tautau is non-fiducial; kept in the signal template it makes the sideband mu-dependent. As a background it costs a 5 % normalisation NP and nothing else.
6. The BDT categories give a nearly background-free Z peak (S/B 10.9) and confine the fake uncertainties (7 %, now honest) to where the fakes are; the statistical gain is small (the measurement is systematics limited).
8. The Tight working point (v3 nominal) vs Medium (v2.1), same chain: μ_Z 1.071 +0.114 -0.100 vs 1.205 +0.145 -0.125, total +11/-10 % vs +14/-12 %: every uncertainty group shrinks, the visible-pT slope is weaker but still there.
7. Reproducibility: this checkout's TRExFitter was a v1.10 commit built against the StatAnalysis ROOT; it is now the tagged v1.8.0 built with fitting/build_trexfitter.sh.

For the combination

workspace fit/results/ztautau (job ztautau, POI mu_Z, regions tautau_SR0/1/2) plugs into fitting/combination_skeleton.config

correlated: Lumi, Pileup, L1Prefiring, QCDScale, PDF, PS_ISR/FSR, XS_*; tau-only: TauID/Trigger/ES_DM*, Fake*_tautau, XS_DYtautau_nonfid

SigModel_tautau is reported, not fitted; never correlate it with z-mumu's SigModel (a different quantity)

with ee/mumu fixing mu_Z, tautau constrains the tau ID nuisance parameters in situ (lepton universality): present TauID_DM* as a result

Born-level 60-120 GeV denominator for sigma; acceptance uncertainty 3.7 % correlated

Next steps

(CLAUDE.md, open issues)

decay-mode categories -> tau ID in situ even before the combination; quote the pT-binned POG prescription as a cross-check

trigger efficiency from mu tau_h tag-and-probe (SingleMuon) instead of the POG turn-on SFs

HT-binned W+jets; the high-mass DY sample for the non-fiducial template; EWK Z->tautau

add e tau_h / mu tau_h channels

Everything here is reproducible from the ntuples in ~25 min: python run_all.py --from 3 (z-tautau/README.md)